covid | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

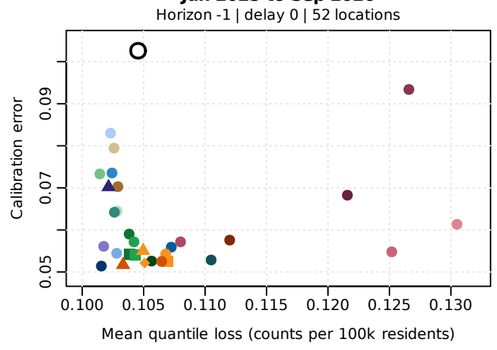
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

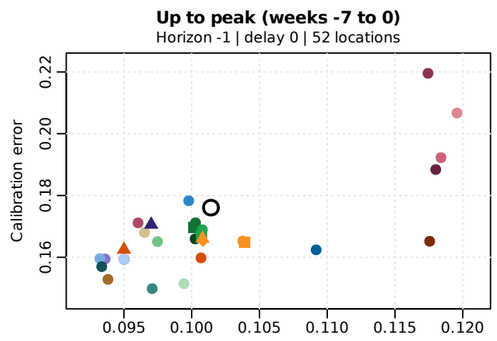
covid | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

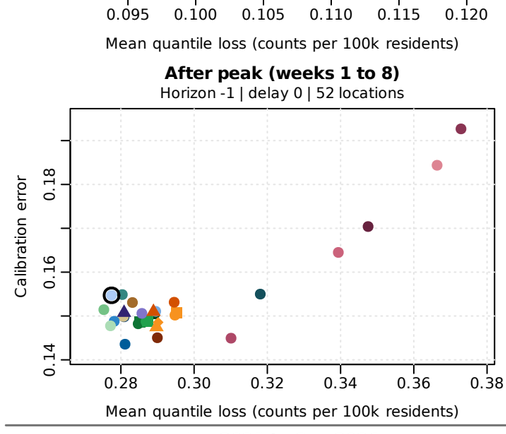
Means over locations only; axes fitted to the means.

Feedback delay 0

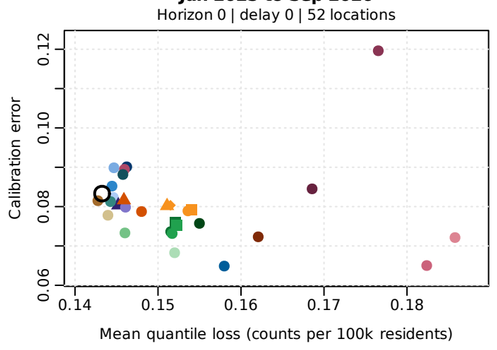
Horizon -1

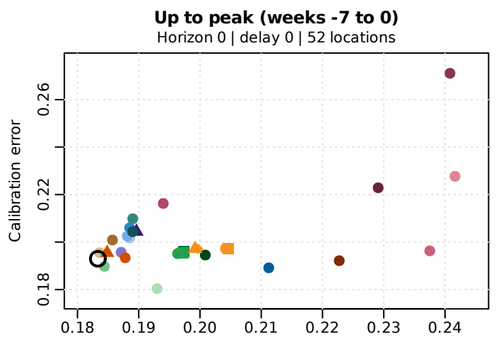


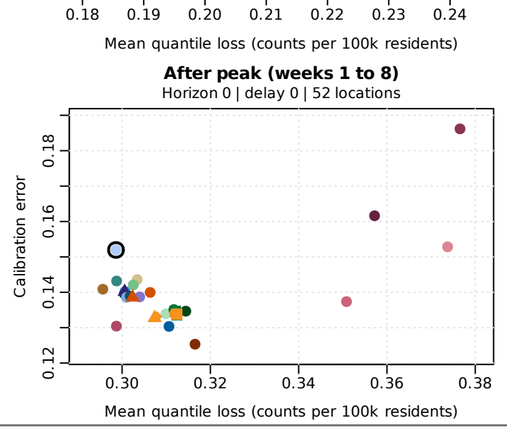




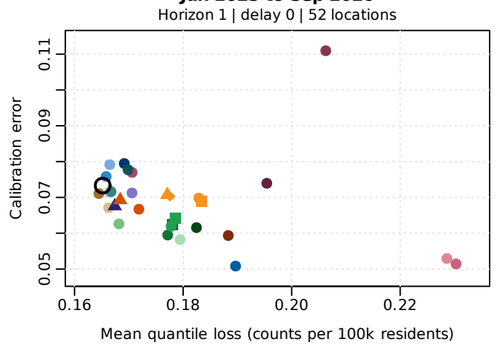
Horizon 0

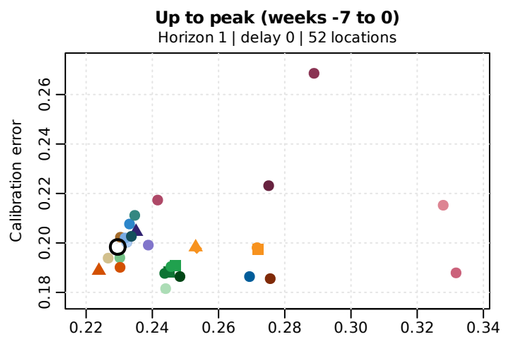


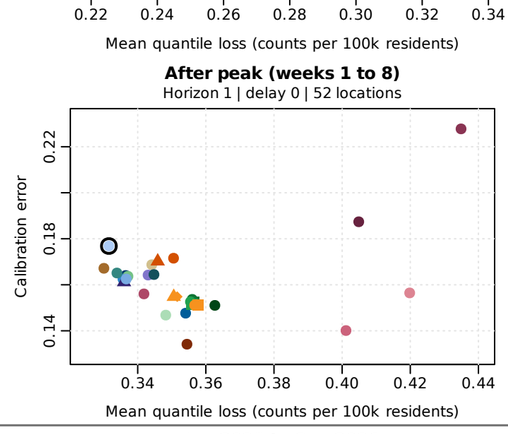




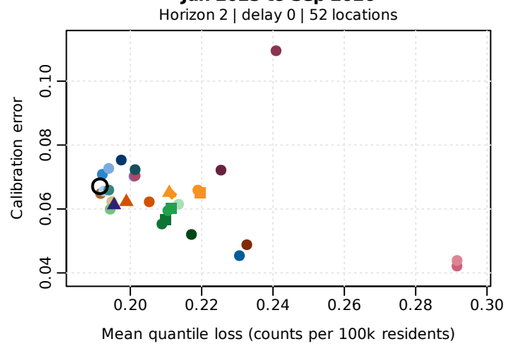
Horizon 1

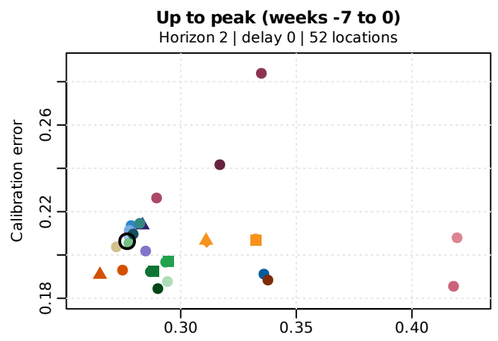


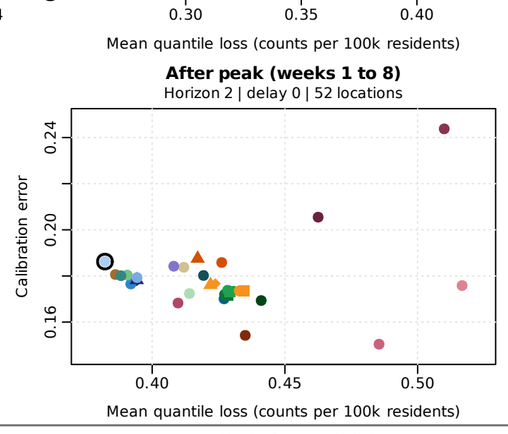




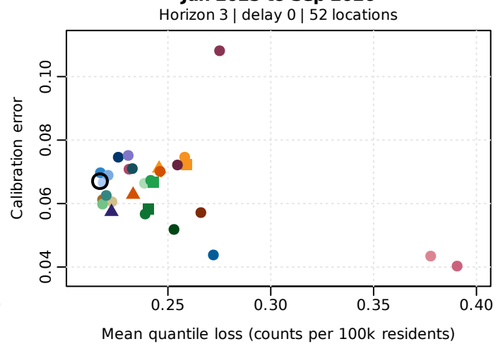
Horizon 2

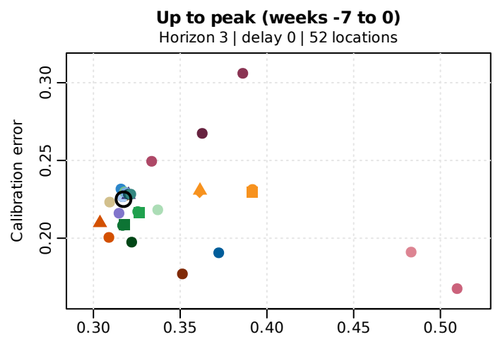


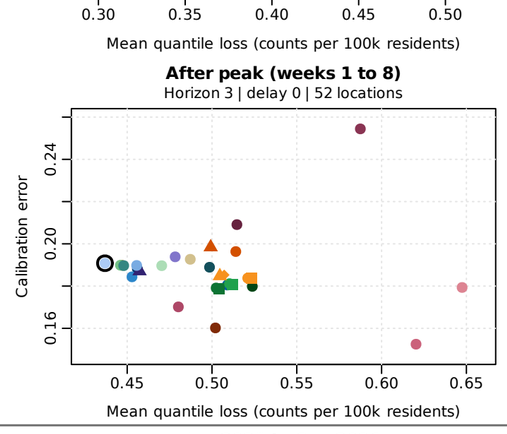




Horizon 3

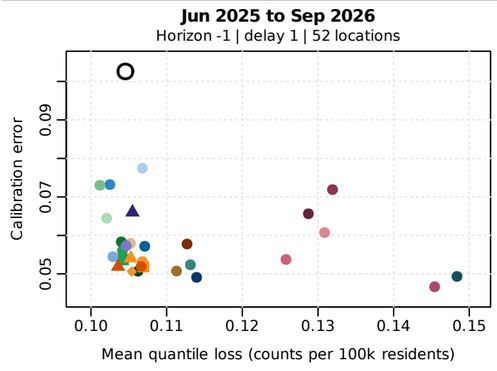


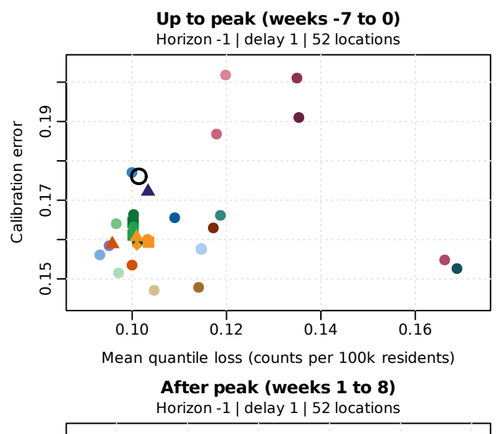


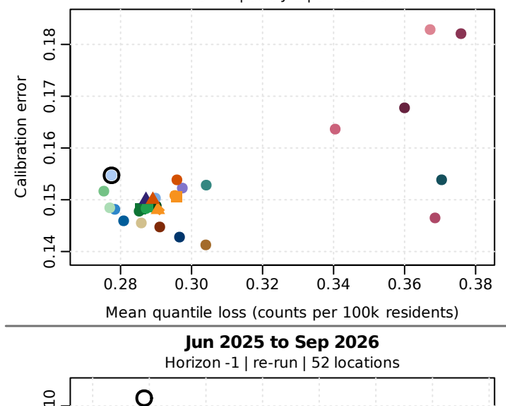


Feedback delay 1

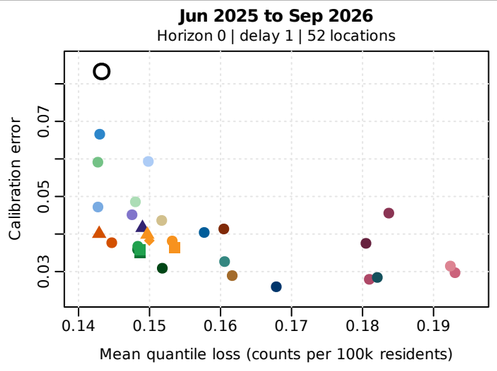
Horizon -1

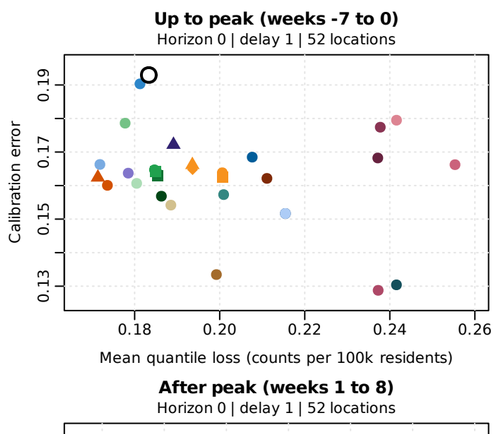


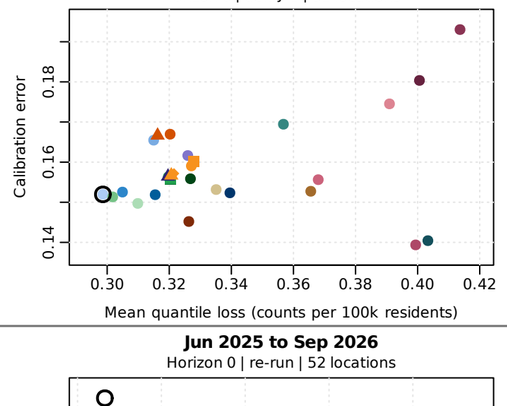




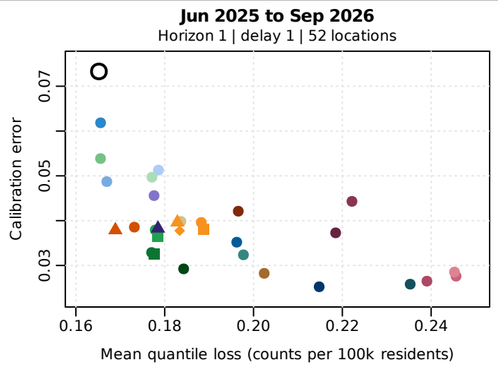
Horizon 0

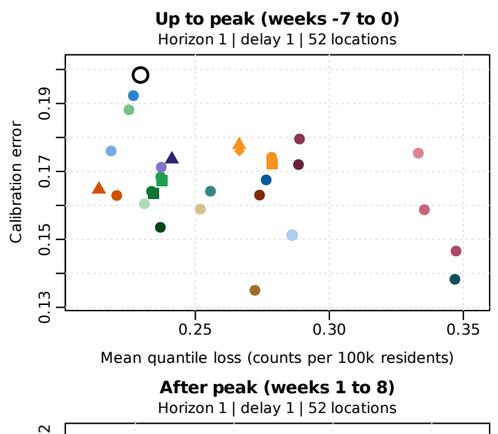


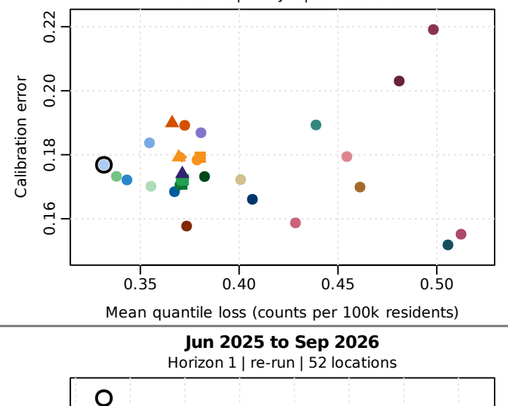




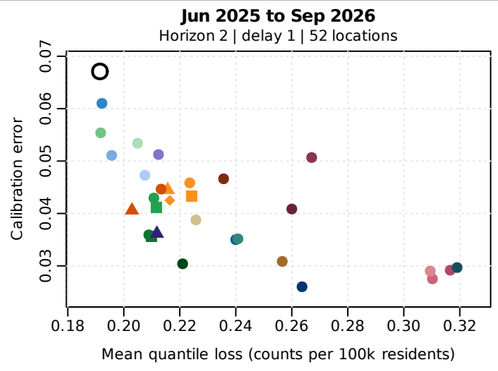
Horizon 1

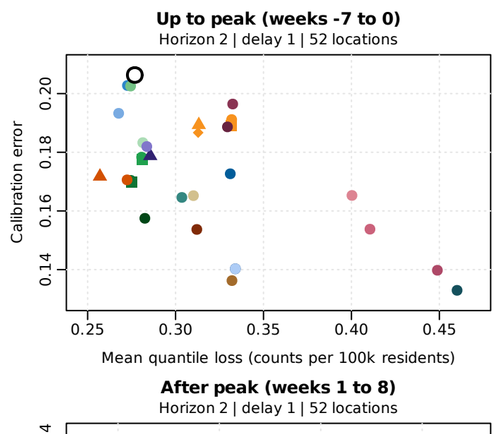


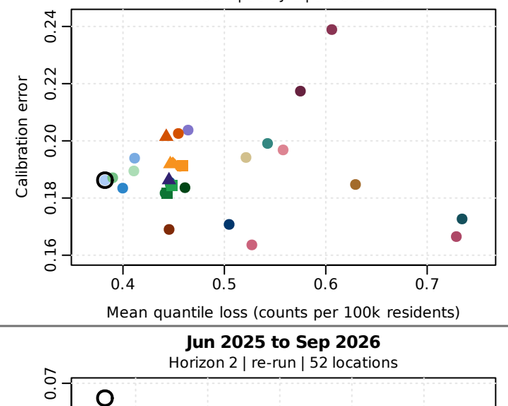




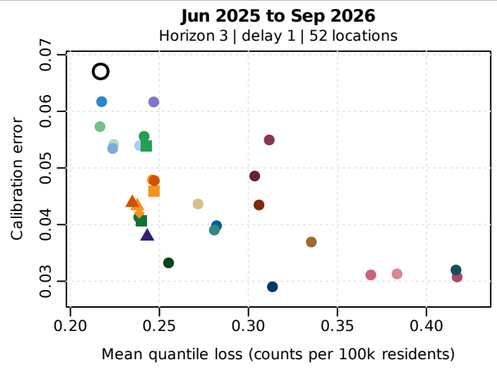
Horizon 2

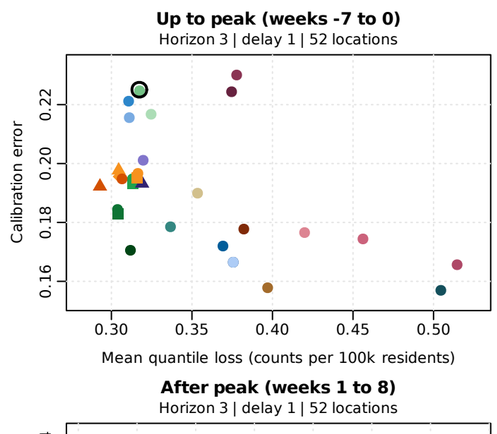


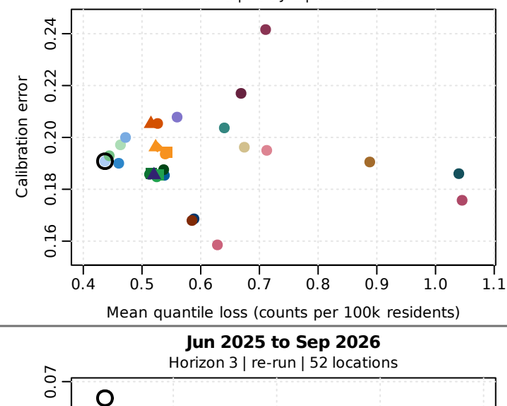




Horizon 3

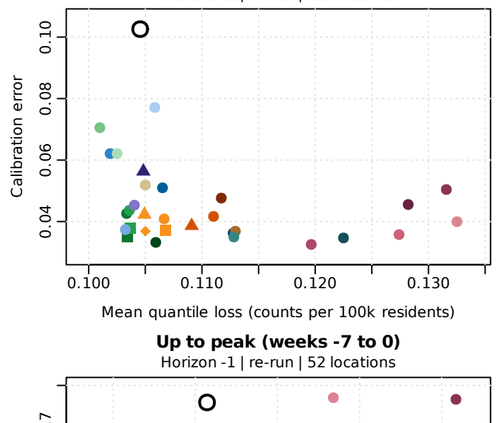


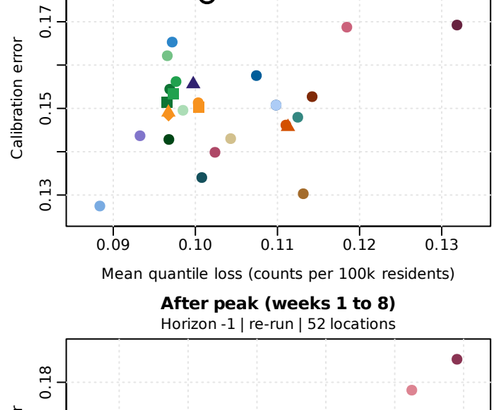


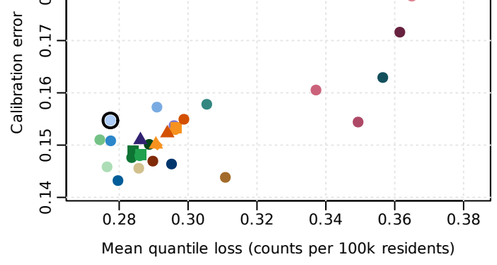


Feedback re-run

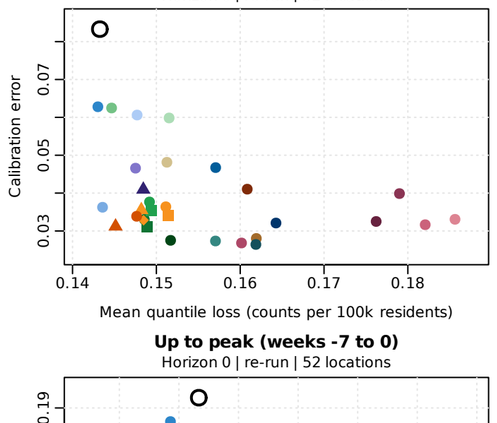
Horizon -1

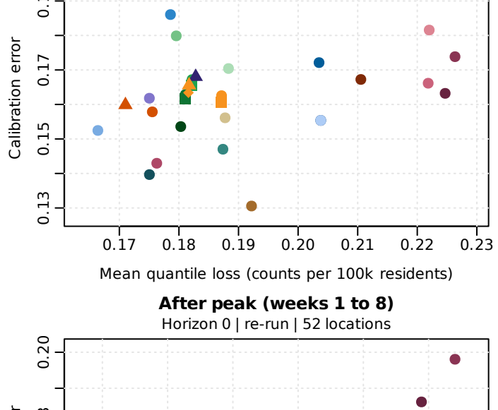


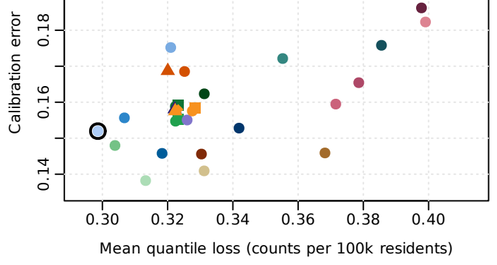




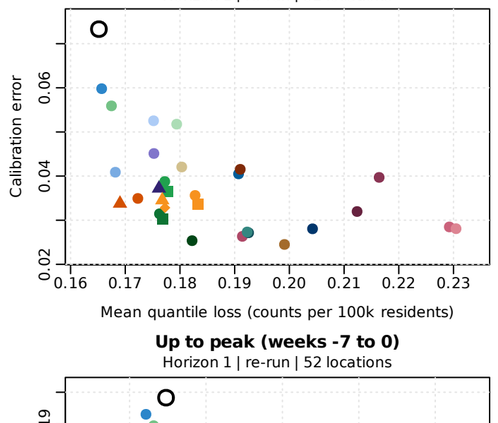
Horizon 0

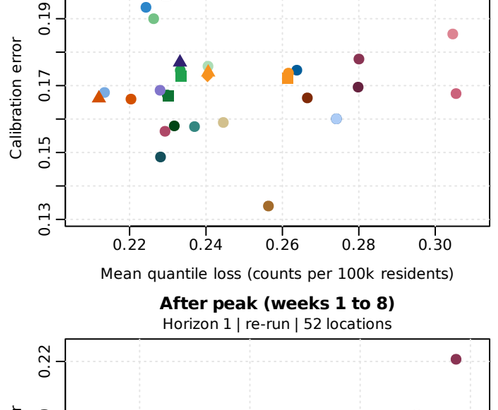


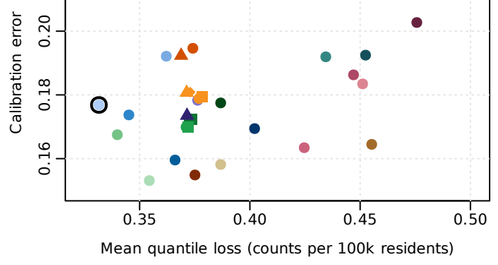




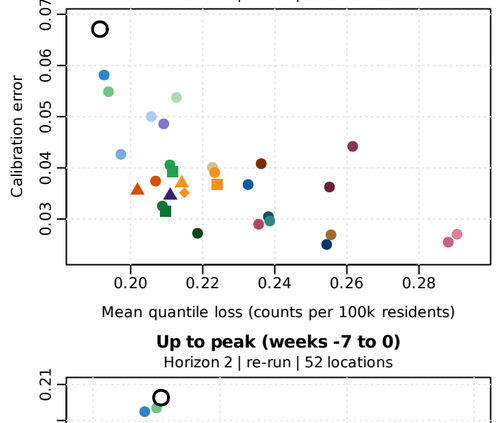
Horizon 1

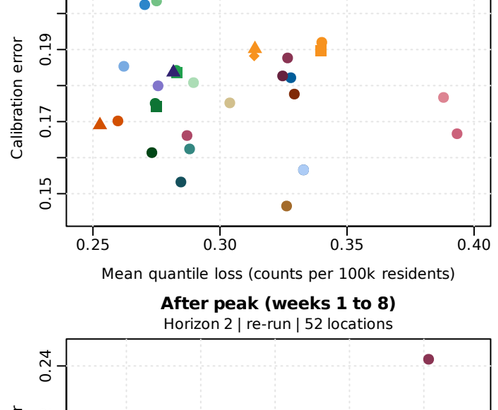


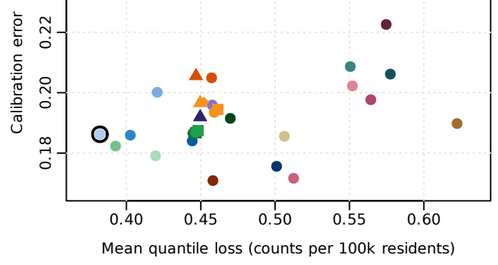




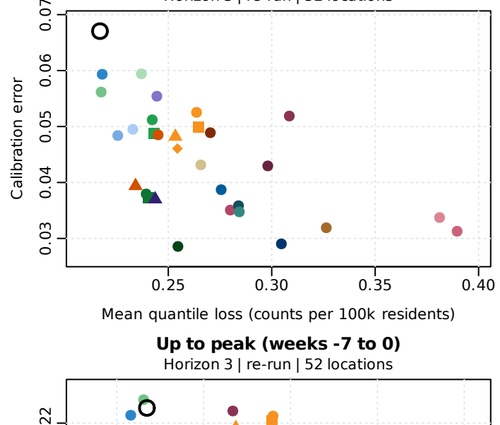
Horizon 2

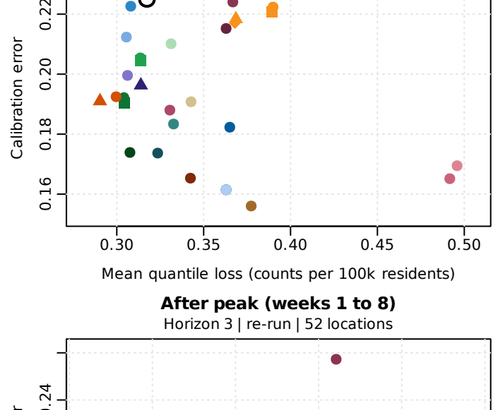


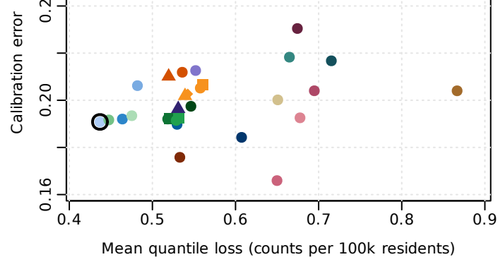




Horizon 3







Single trackers
- Base
- Fixed MultiQT (eta=0.01*S0)
- Leaky MultiQT (eta=0.1*S0, leak 0.1)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Target oracle (hindsight)

Log1p tracker
- Fixed MultiQT, log1p tracker (eta=0.01*S0)
- Leaky MultiQT, log1p tracker (eta=0.05*S0, leak 0.1)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=8)
- Rate grid: learned, no sharing, common weights (H=8)
- Rate grid: learned + sharing (H=8)
- Rate grid: learned + sharing, common weights (H=8)
- Rate grid: sharing + outer (H=8, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=8, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
- Starts equal + outer (outer m=0.001)
- Starts learned + sharing + outer (H=8, outer m=0.001)

Score-based experts
- Score experts: equal weights
- Score experts: learned weights per level (H=8)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.1)
- Studentized median (RMS of 5, median start, eta=0.1)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, median start, eta=0.3)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

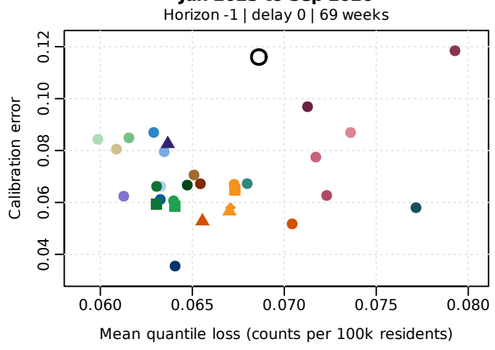
covid | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

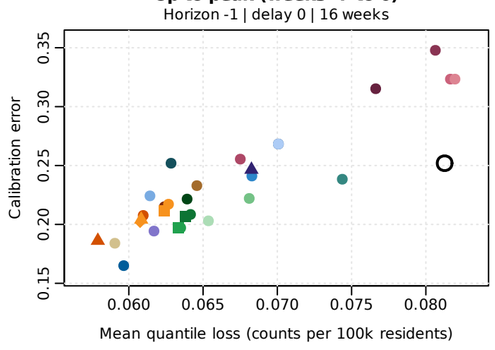
Feedback delay 0

Horizon -1

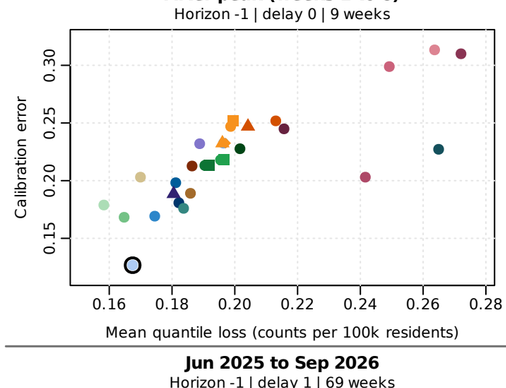
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

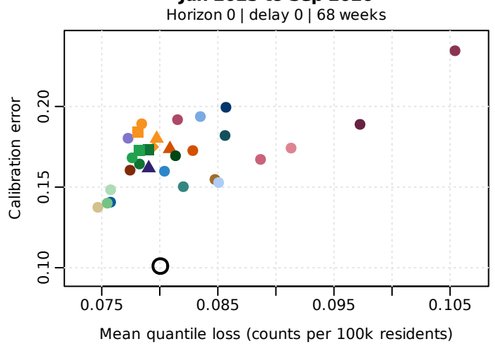


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 9 weeks

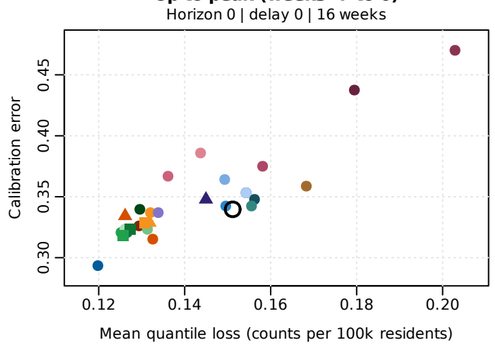


Horizon 0

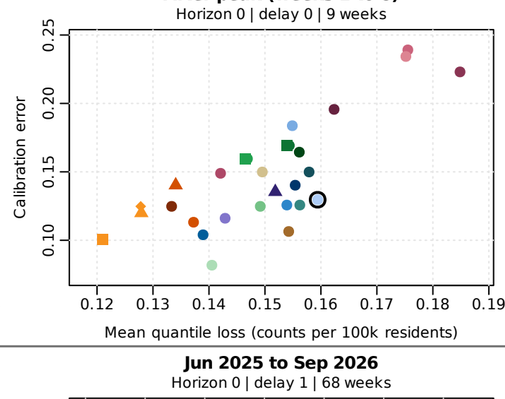
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 16 weeks

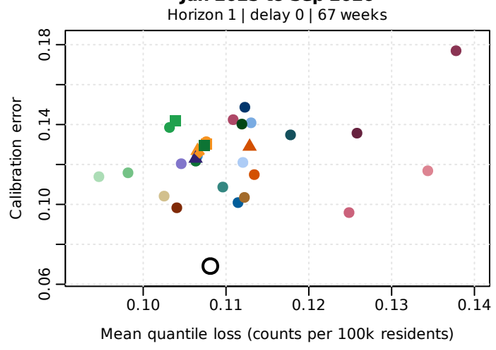


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 9 weeks

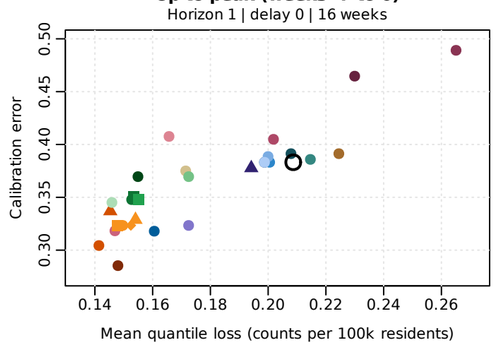


Horizon 1

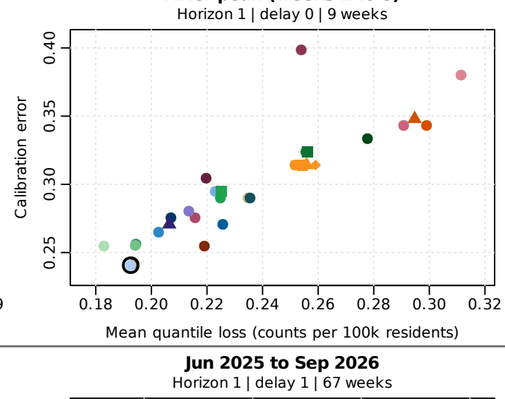
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 16 weeks

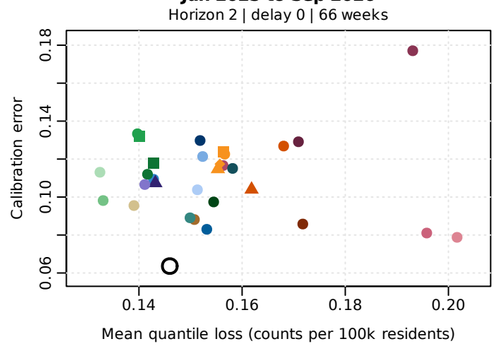


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 9 weeks

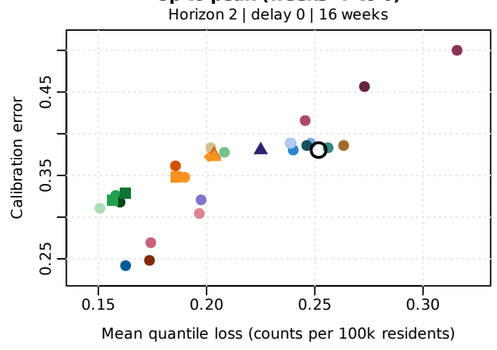


Horizon 2

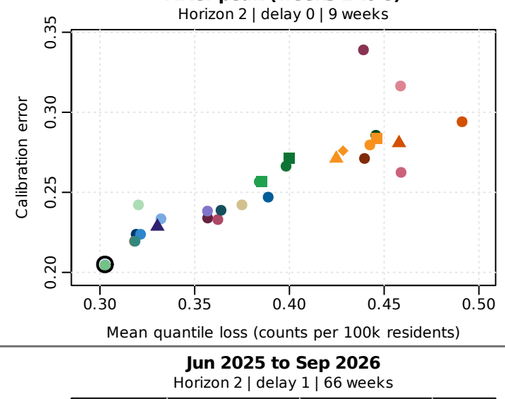
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 16 weeks

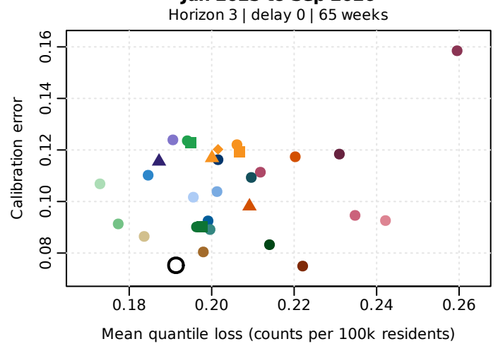


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 9 weeks

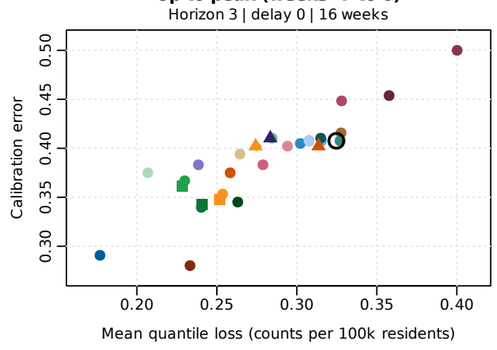


Horizon 3

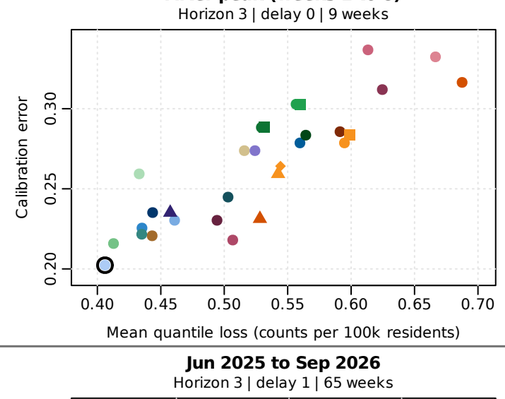
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 16 weeks



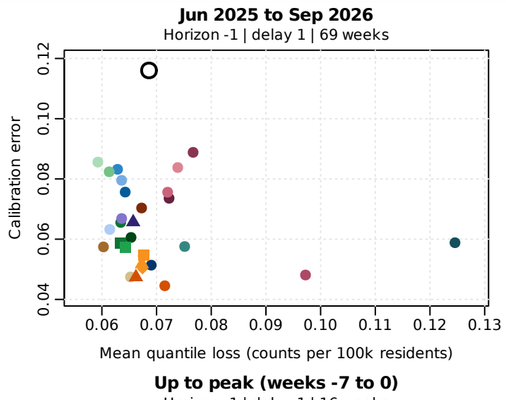
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 9 weeks



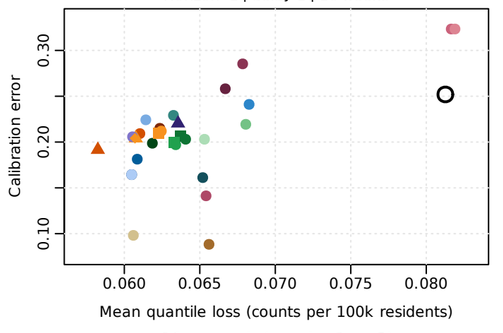
Feedback delay 1

Horizon -1

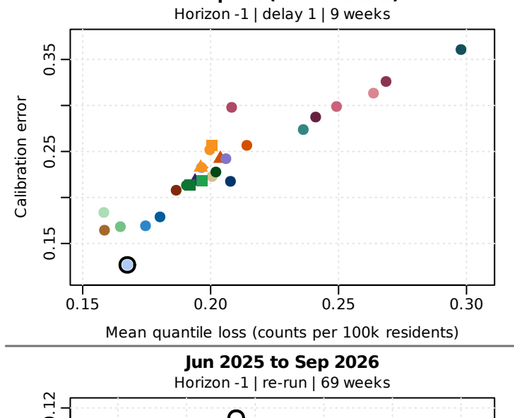
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

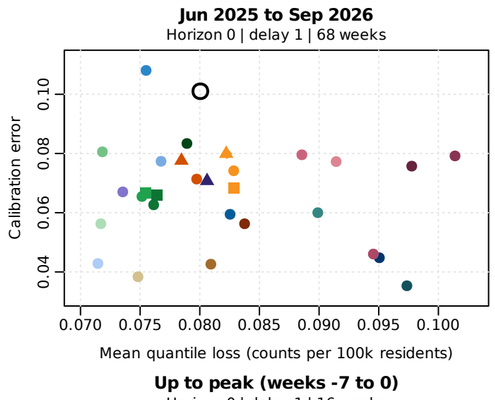


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 9 weeks

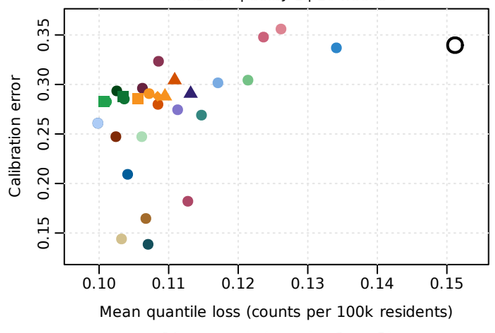


Horizon 0

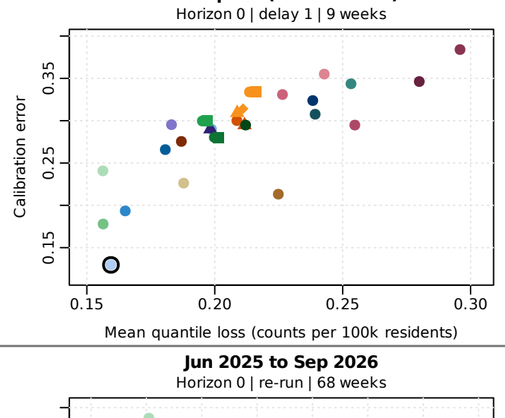
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 16 weeks

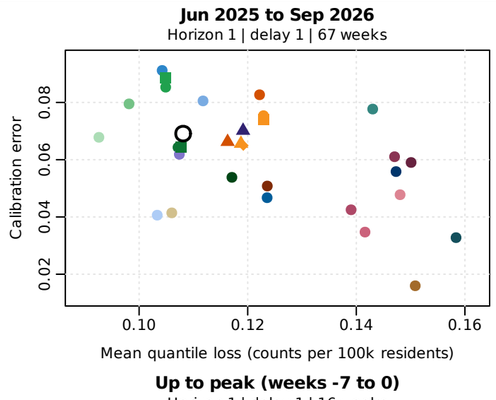


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 9 weeks

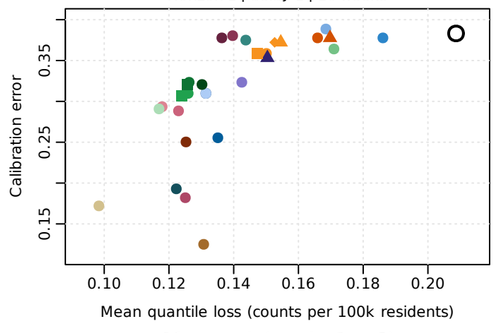


Horizon 1

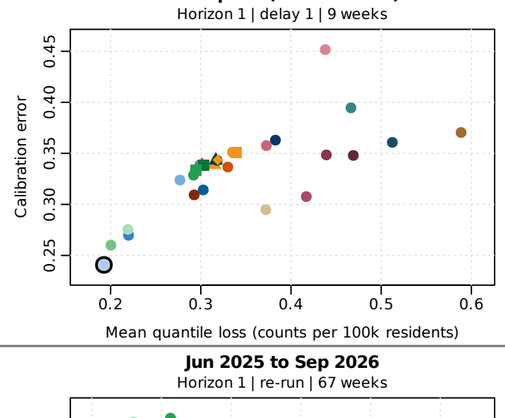
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 16 weeks

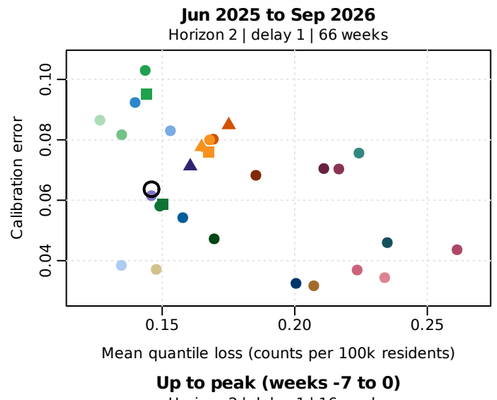


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 9 weeks

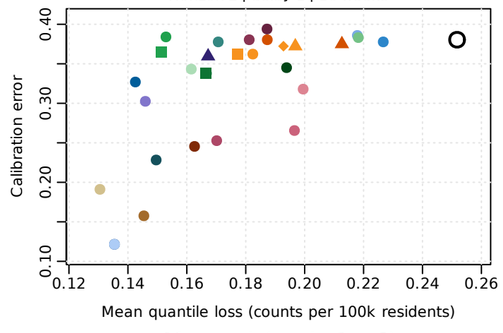


Horizon 2

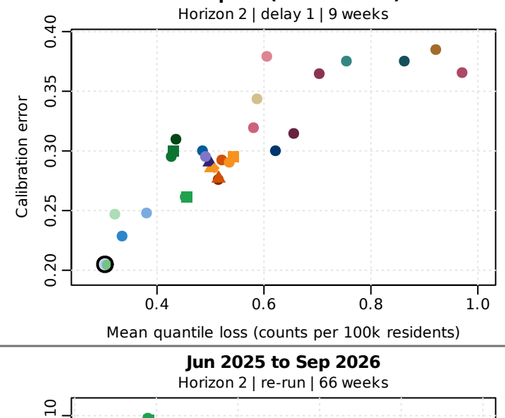
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 16 weeks

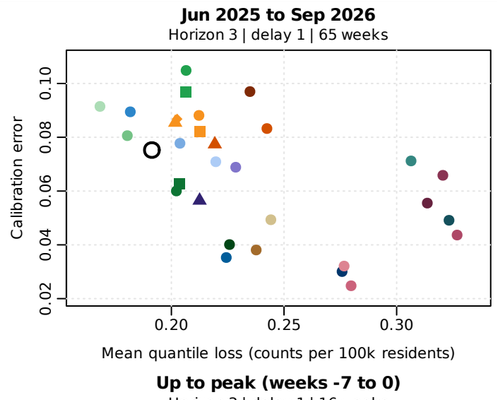


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 9 weeks

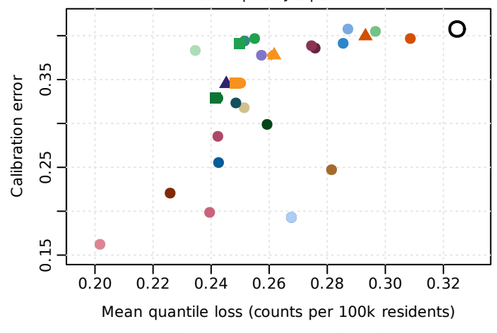


Horizon 3

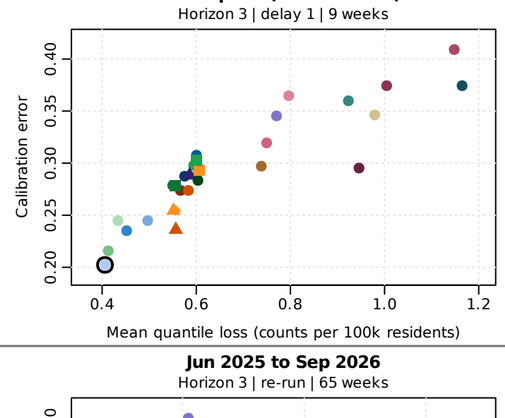
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 16 weeks



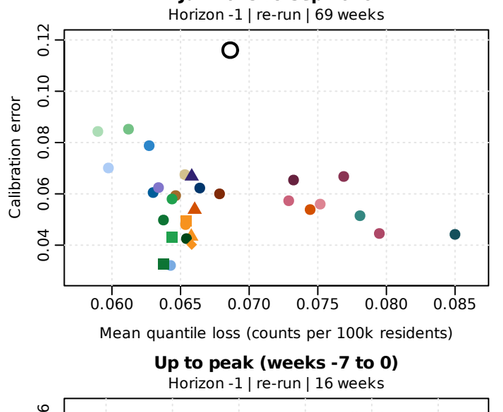
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 9 weeks



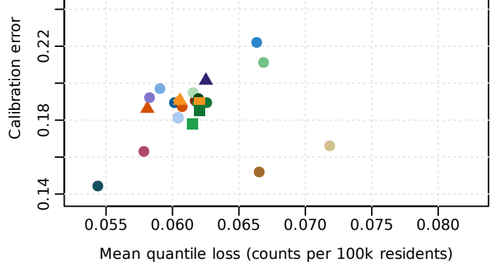
Feedback re-run

Horizon -1

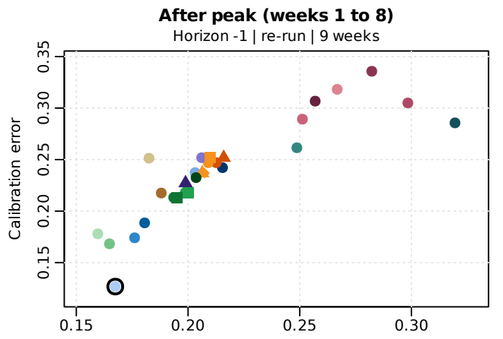
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks

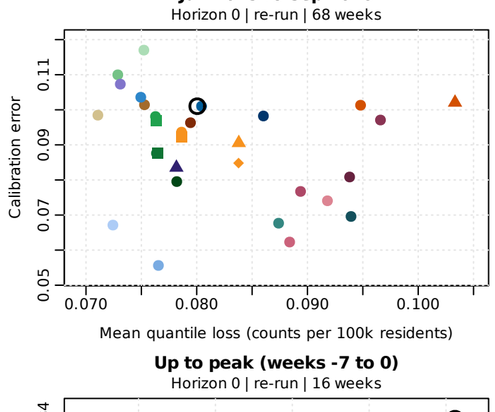


After peak (weeks 1 to 8)
Horizon -1 | re-run | 9 weeks

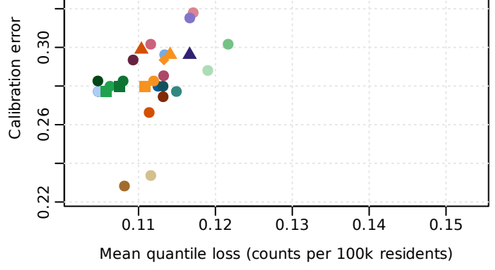


Horizon 0

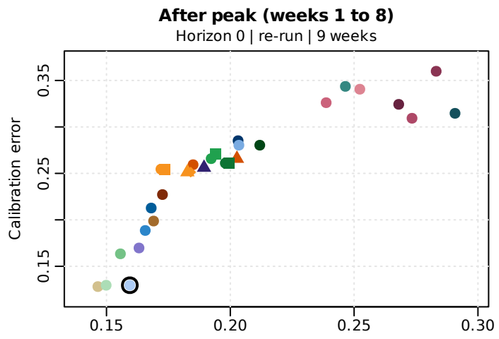
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 16 weeks

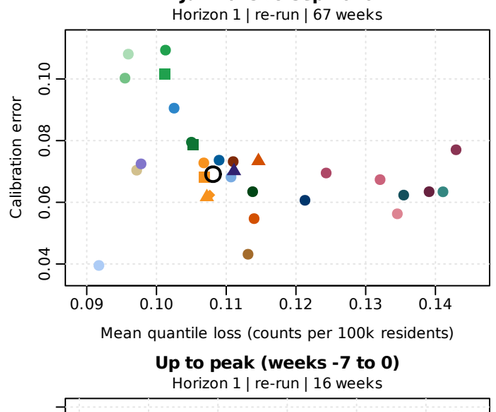


After peak (weeks 1 to 8)
Horizon 0 | re-run | 9 weeks

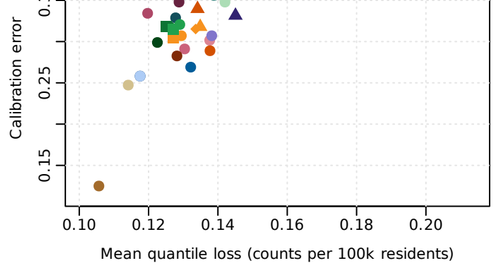


Horizon 1

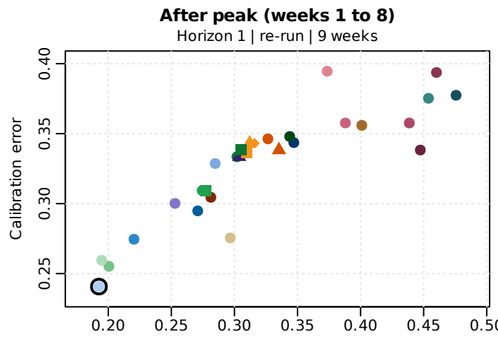
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Up to peak (weeks -7 to 0)
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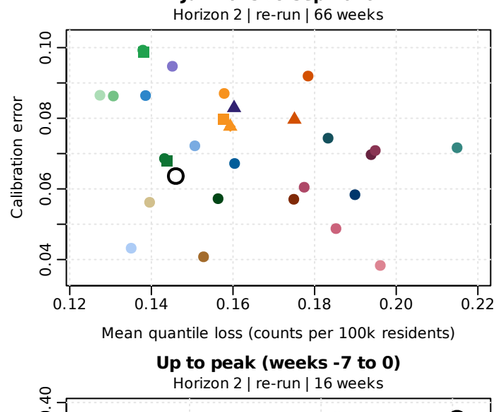


After peak (weeks 1 to 8)
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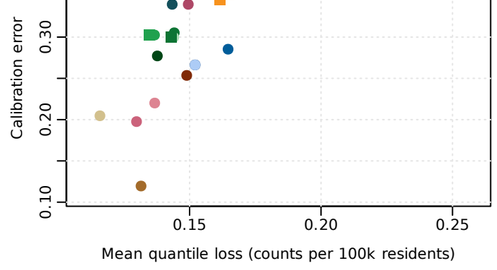


Horizon 2

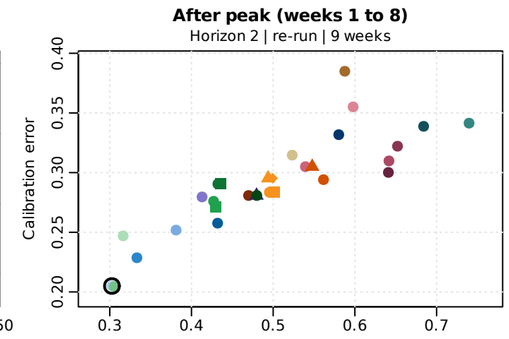
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 16 weeks

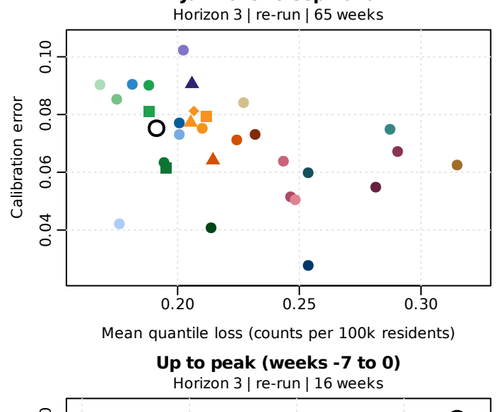


After peak (weeks 1 to 8)
Horizon 2 | re-run | 9 weeks

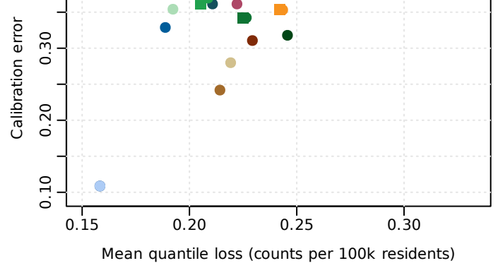


Horizon 3

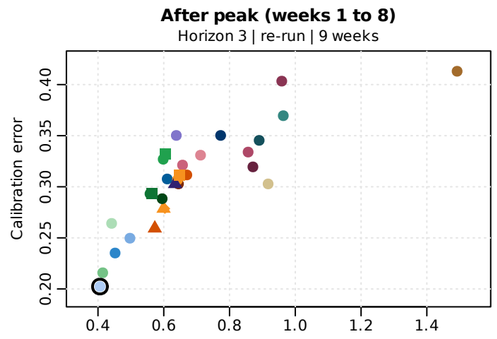
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 16 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 9 weeks



Single trackers
- Base
- Fixed MultiQT (eta=0.01*S0)
- Leaky MultiQT (eta=0.1*S0, leak 0.1)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Target oracle (hindsight)

Log1p tracker
- Fixed MultiQT, log1p tracker (eta=0.01*S0)
- Leaky MultiQT, log1p tracker (eta=0.05*S0, leak 0.1)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=8)
- Rate grid: learned, no sharing, common weights (H=8)
- Rate grid: learned + sharing (H=8)
- Rate grid: learned + sharing, common weights (H=8)
- Rate grid: sharing + outer (H=8, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=8, outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
- Starts equal + outer (outer m=0.001)
- Starts learned + sharing + outer (H=8, outer m=0.001)

Score-based experts
- Score experts: equal weights
- Score experts: learned weights per level (H=8)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.1)
- Studentized median (RMS of 5, median start, eta=0.1)
- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, median start, eta=0.3)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

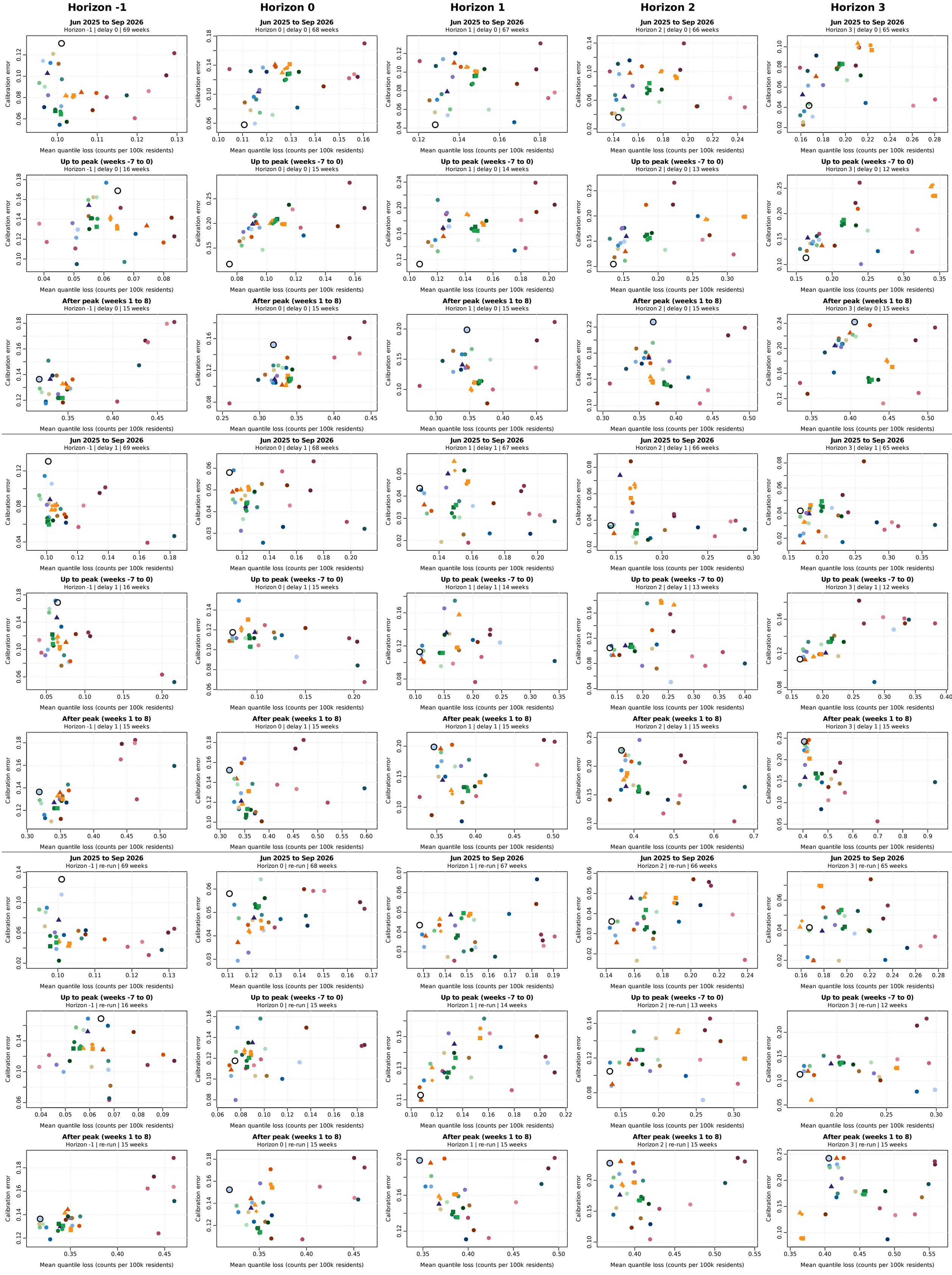
covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.



Single trackers
- Base
- Fixed MultiQT (eta=0.01*S0)
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- Target oracle (hindsight)

Log1p tracker
- Fixed MultiQT, log1p tracker (eta=0.01*S0)
- Leaky MultiQT, log1p tracker (eta=0.05*S0, leak 0.1)

Rate-grid experts
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- Rate grid: learned + sharing (H=8)
- Rate grid: learned + sharing, common weights (H=8)
- Rate grid: sharing + outer (H=8, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=8, outer m=0.001)

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- First start only (m=0.05)
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- Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
- Starts equal + outer (outer m=0.001)
- Starts learned + sharing + outer (H=8, outer m=0.001)

Score-based experts
- Score experts: equal weights
- Score experts: learned weights per level (H=8)

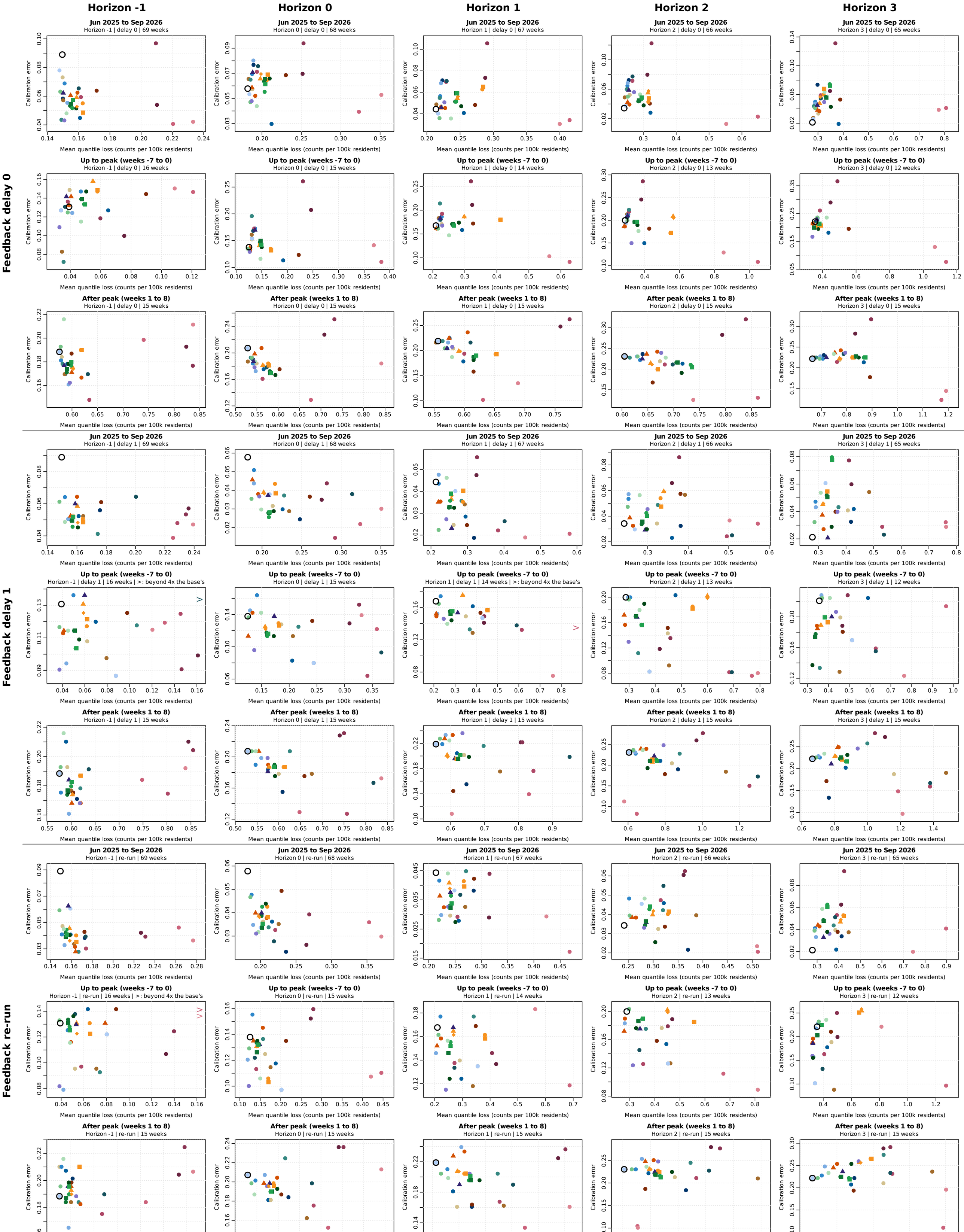
Studentized median
- Studentized median (RMS of 5, normal start, eta=0.1)
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- Studentized median (base 80% width, normal start, eta=0.1)
- Median, constant sigma (RMS of 5, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, median start, eta=0.3)

Studentized full base
- Studentized full base (base 80% width, eta=0.1)
- Studentized full base (level RMS of 5, eta=0.03)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

covid | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

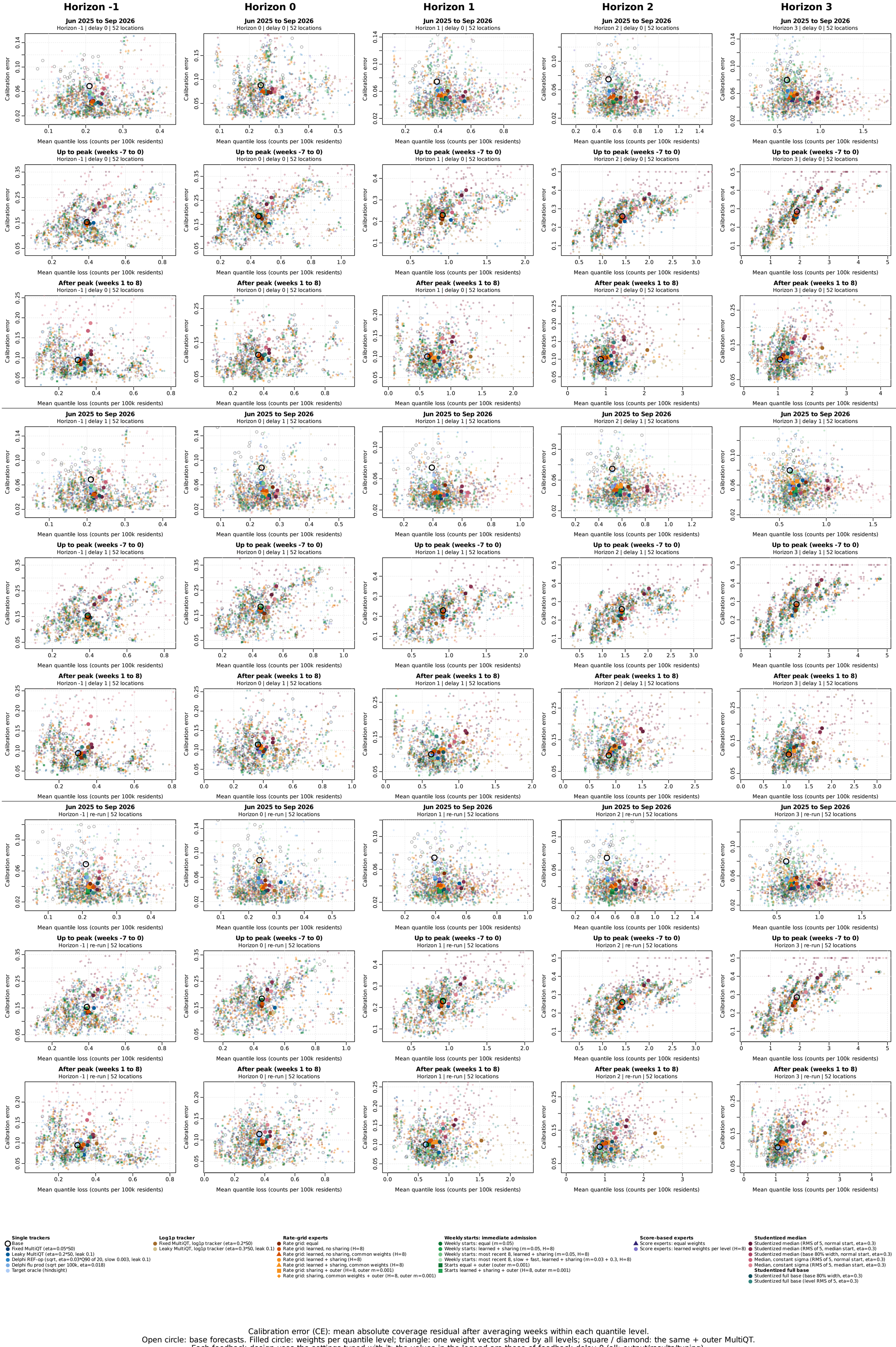
Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

flu | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.



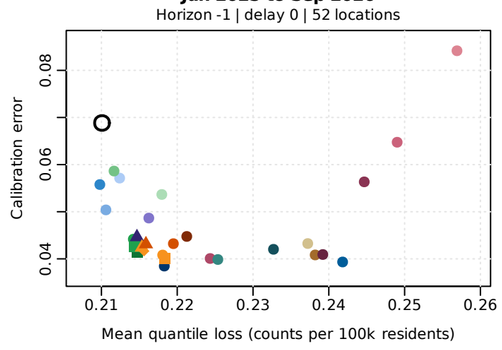
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

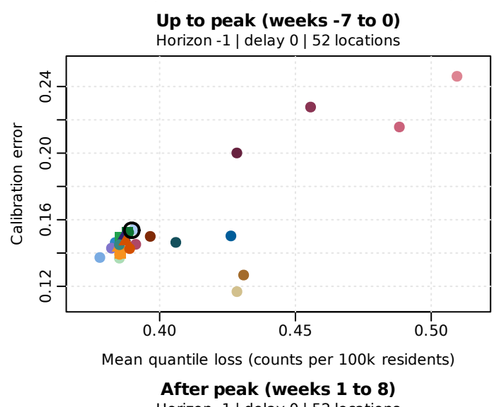
flu | all 52 locations | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

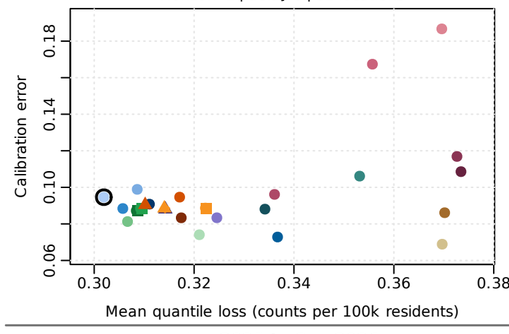
Means over locations only; axes fitted to the means.

Feedback delay 0

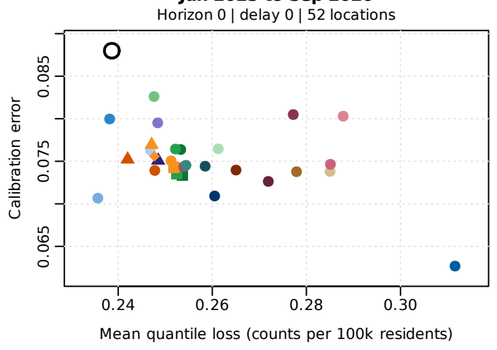
Horizon -1

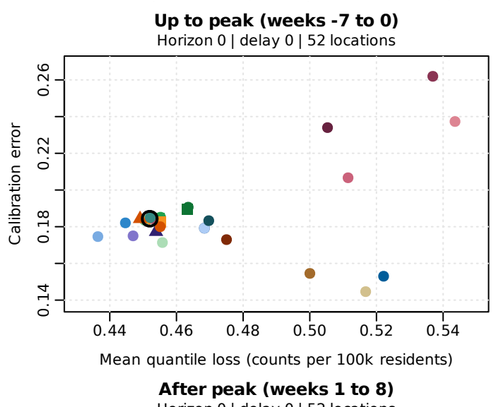


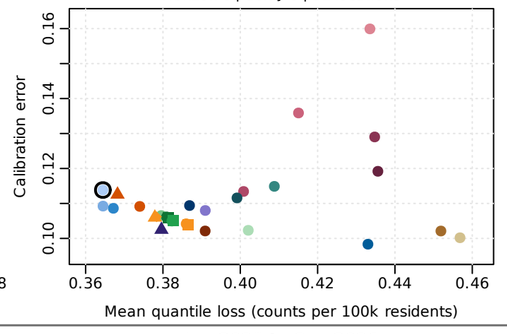




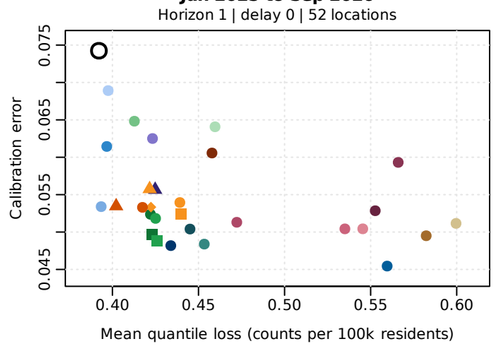
Horizon 0

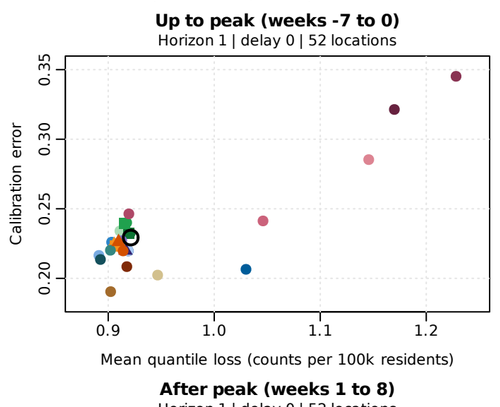


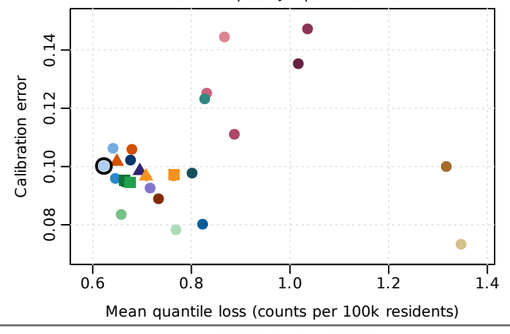




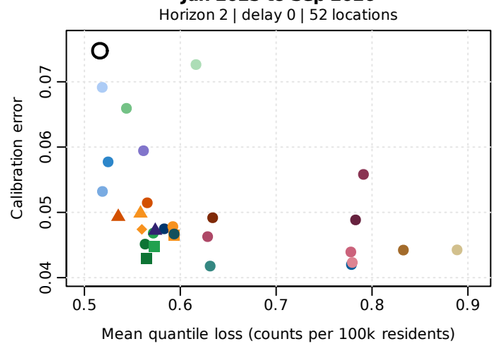
Horizon 1

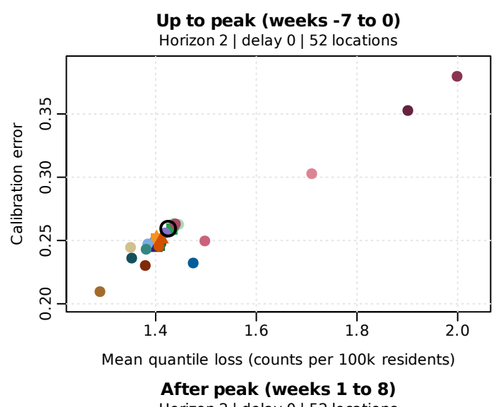


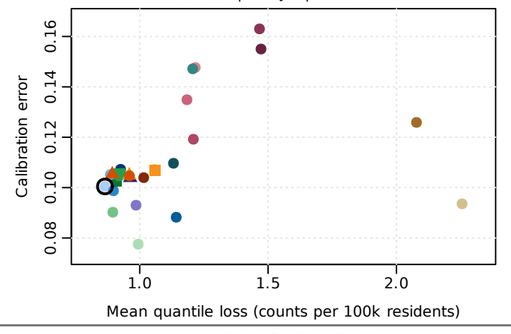




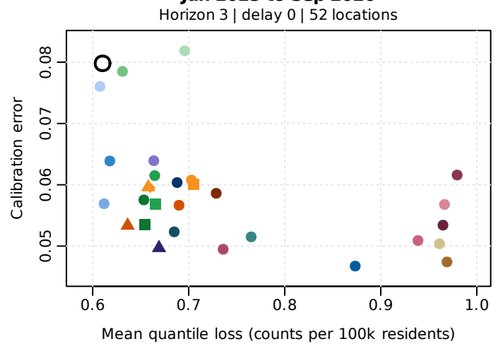
Horizon 2

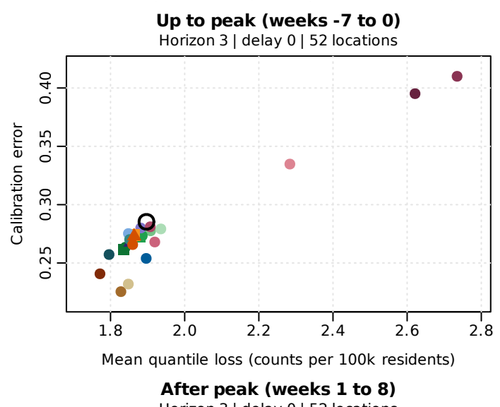


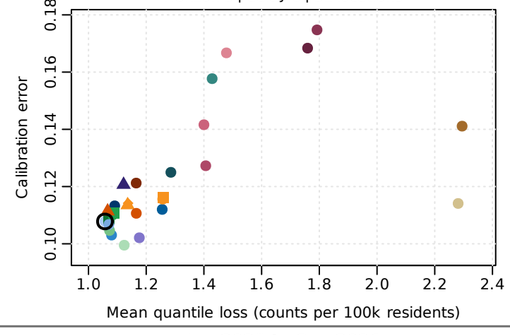




Horizon 3

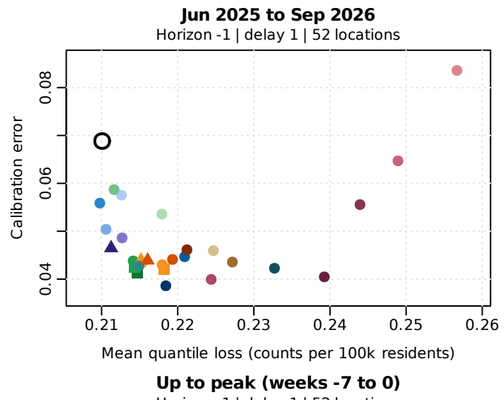


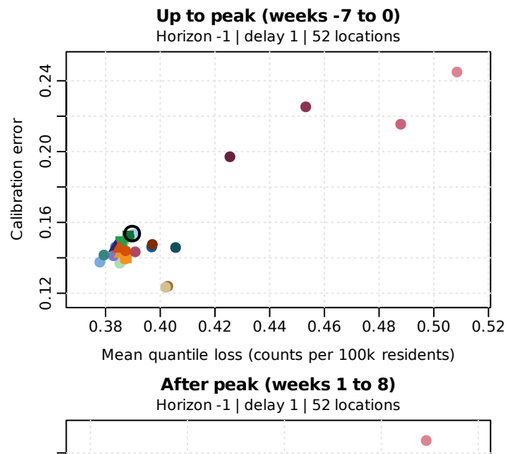


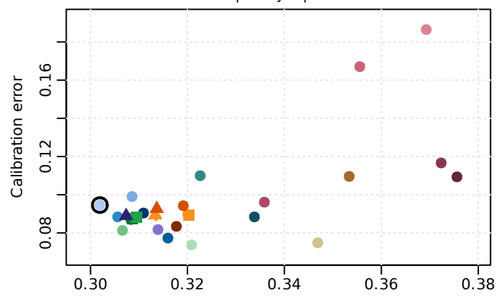


Feedback delay 1

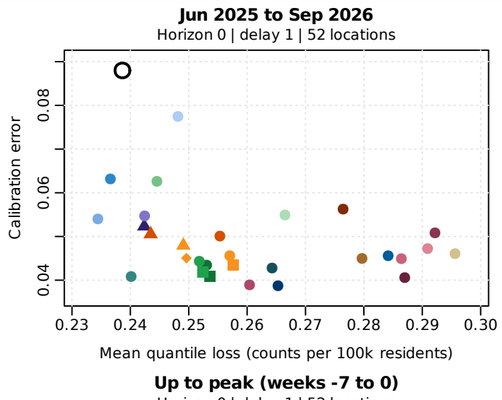
Horizon -1

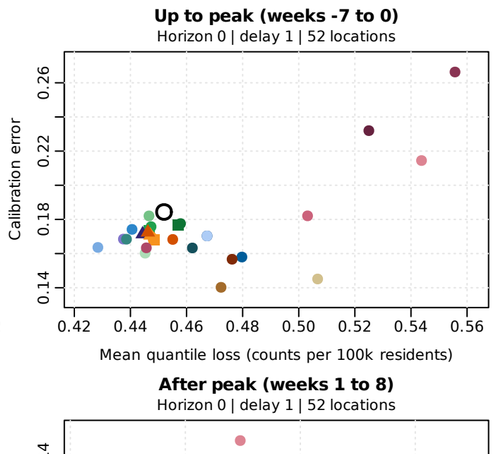


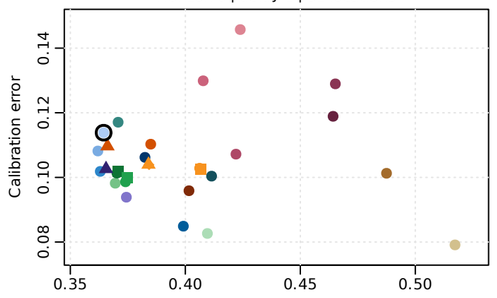




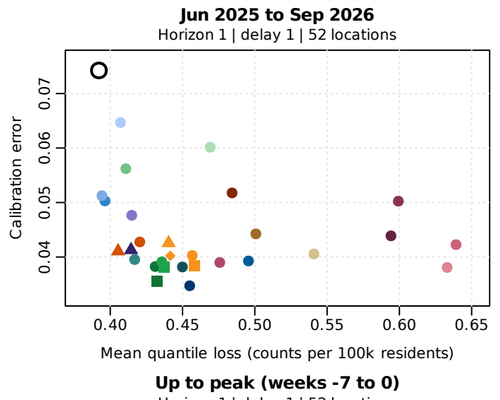
Horizon 0

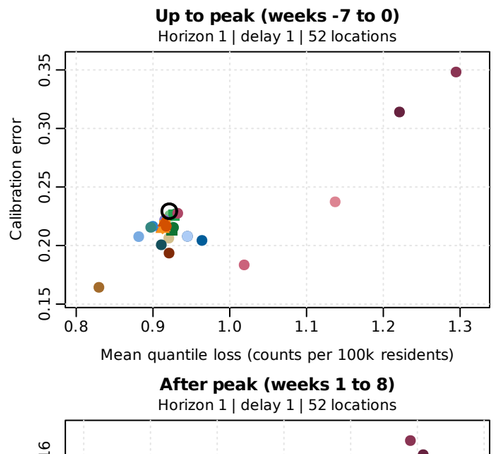


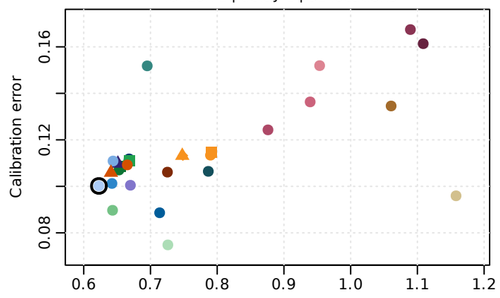




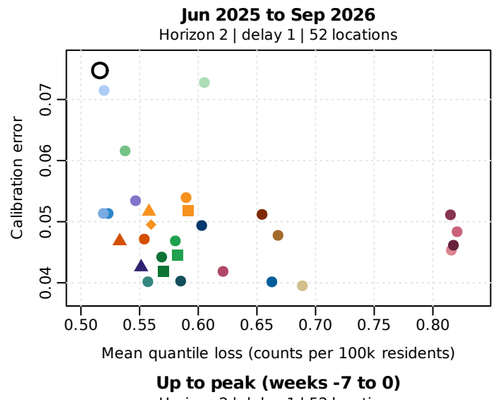
Horizon 1

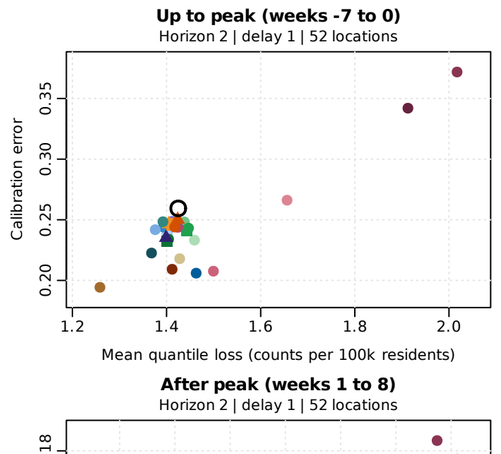


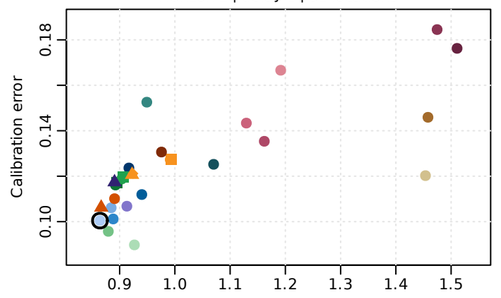




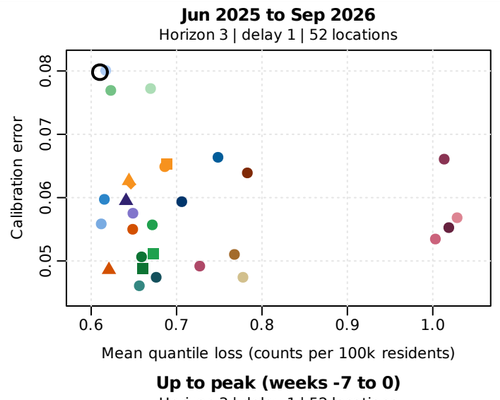
Horizon 2

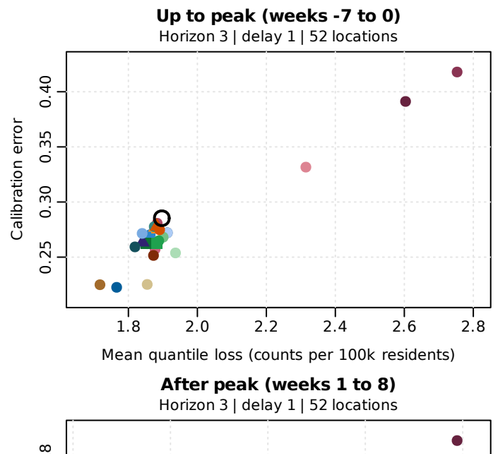


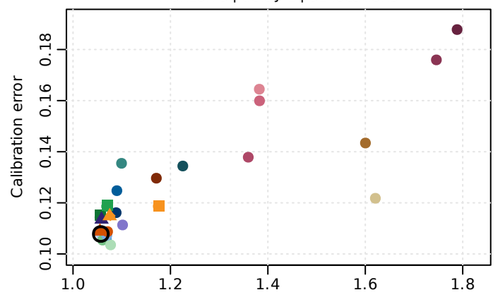




Horizon 3

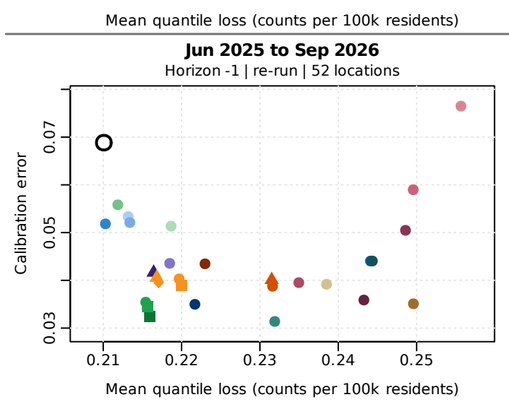


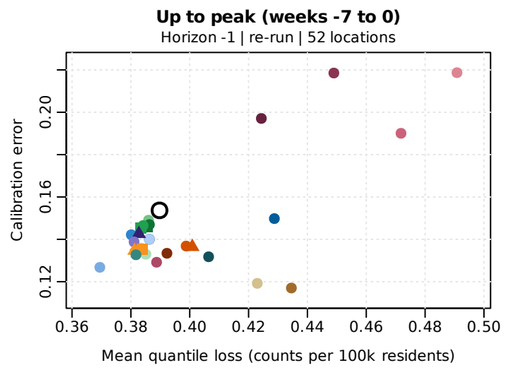


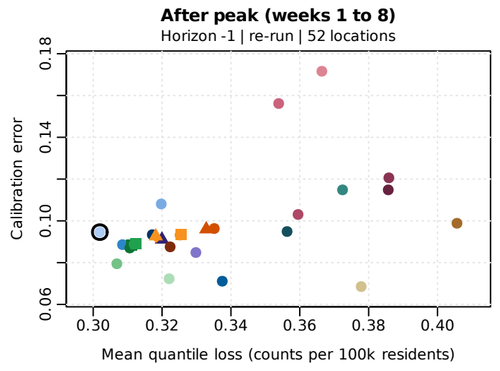


Feedback re-run

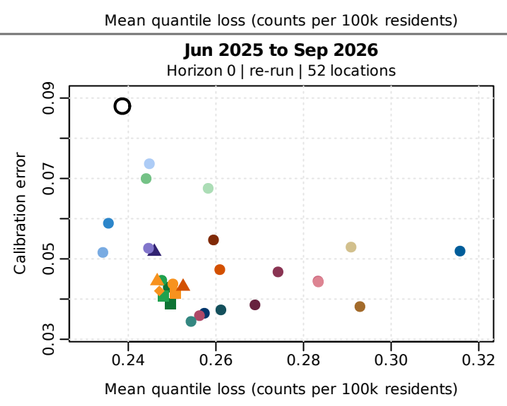
Horizon -1

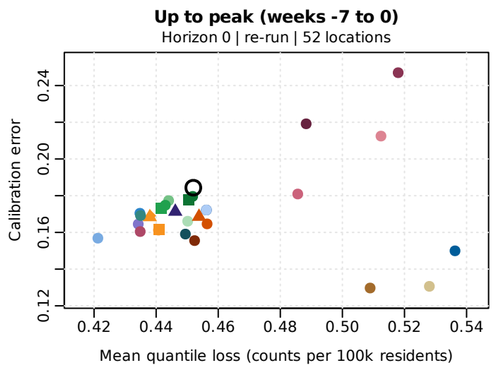


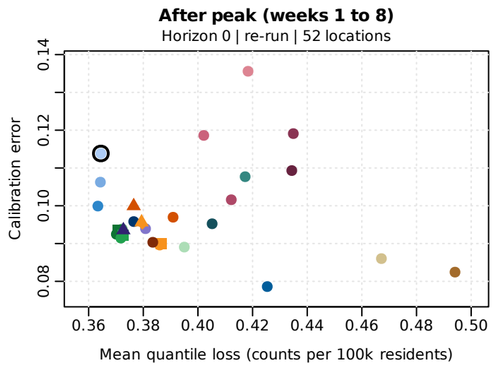




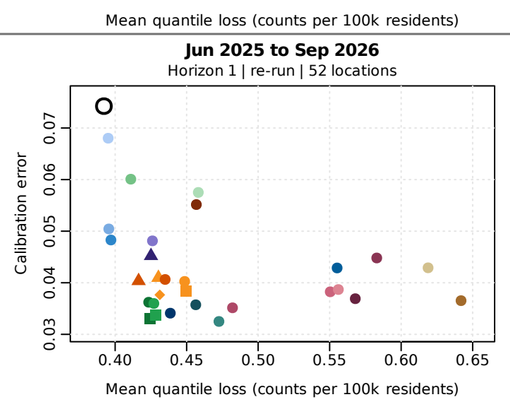
Horizon 0

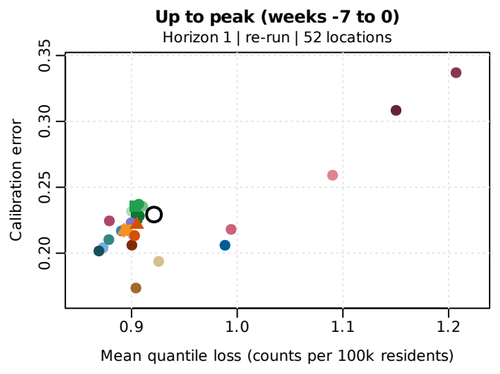


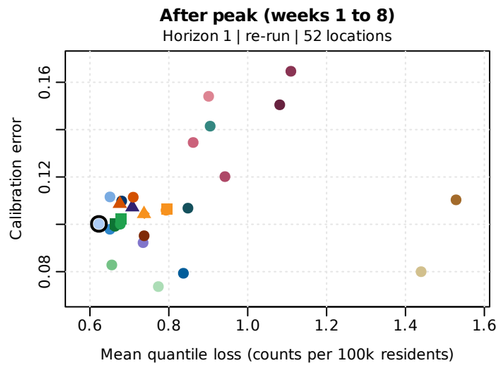




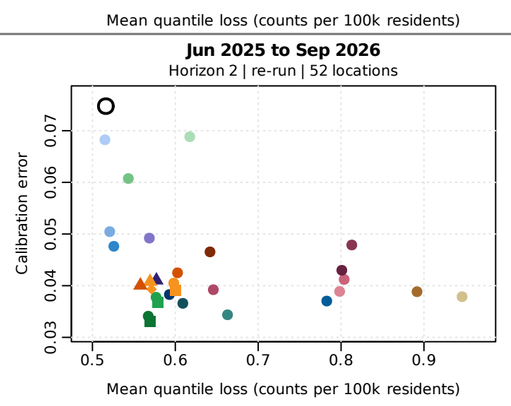
Horizon 1

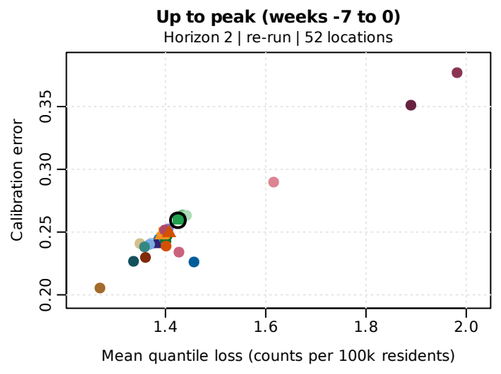


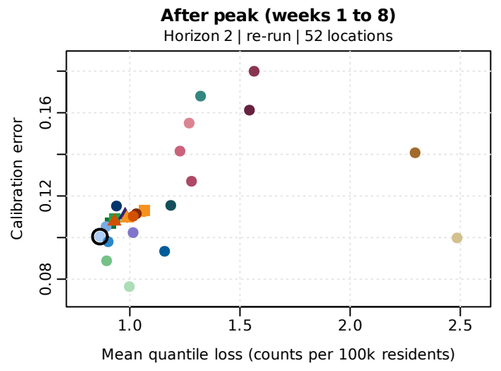




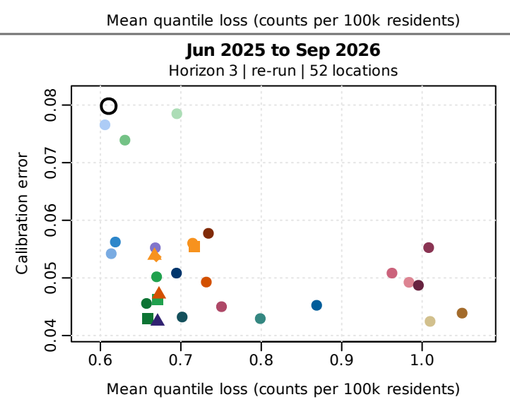
Horizon 2

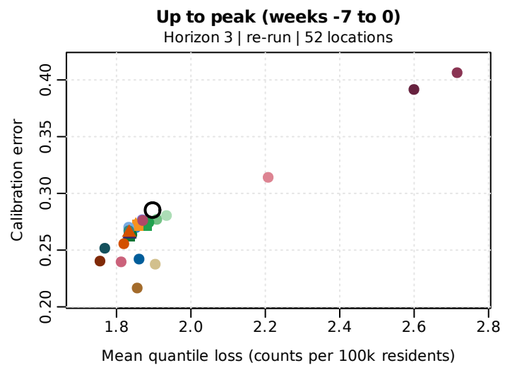


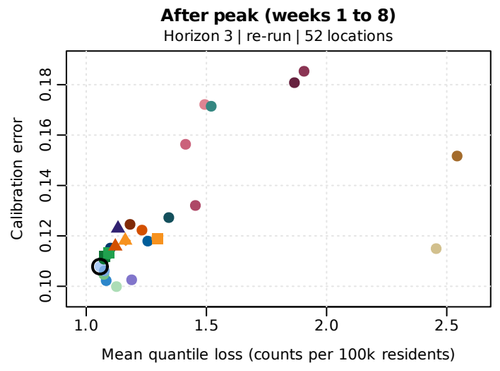




Horizon 3







Single trackers
- Base
- Fixed MultiQT (eta=0.05*S0)
- Leaky MultiQT (eta=0.2*S0, leak 0.1)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Target oracle (hindsight)

Log1p tracker
- Fixed MultiQT, log1p tracker (eta=0.2*S0)
- Leaky MultiQT, log1p tracker (eta=0.3*S0, leak 0.1)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=8)
- Rate grid: learned, no sharing, common weights (H=8)
- Rate grid: learned + sharing (H=8)
- Rate grid: learned + sharing, common weights (H=8)
- Rate grid: sharing + outer (H=8, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=8, outer m=0.001)

Weekly starts: immediate admission
- Weekly starts: equal (m=0.05)
- Weekly starts: learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
- Starts equal + outer (outer m=0.001)
- Starts learned + sharing + outer (H=8, outer m=0.001)

Score-based experts
- Score experts: equal weights
- Score experts: learned weights per level (H=8)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.3)
- Studentized median (RMS of 5, median start, eta=0.3)
- Studentized median (base 80% width, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, median start, eta=0.3)

Studentized full base
- Studentized full base (base 80% width, eta=0.3)
- Studentized full base (level RMS of 5, eta=0.3)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

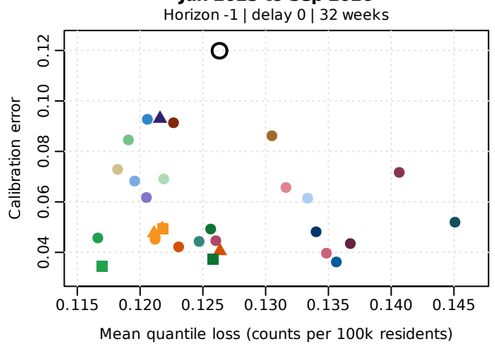
flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

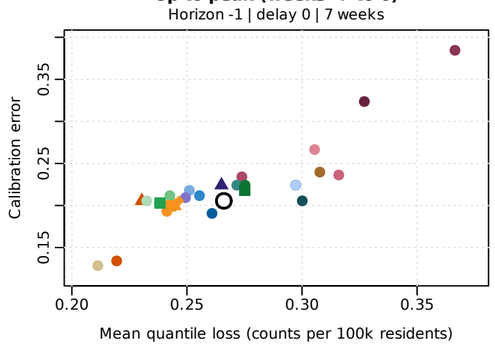
Horizon -1

Feedback delay 0

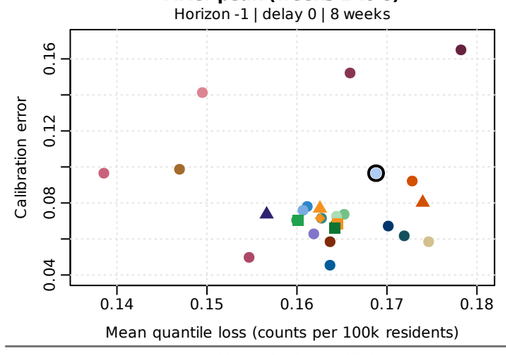
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks

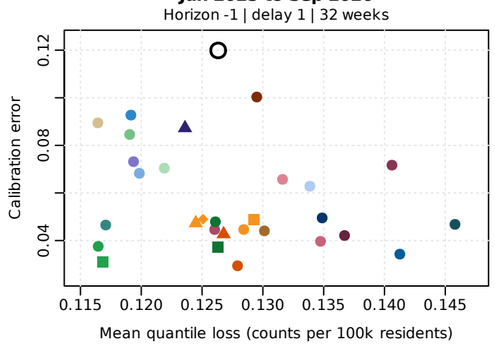


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

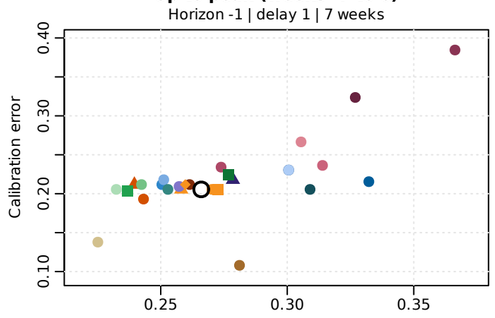


Feedback delay 1

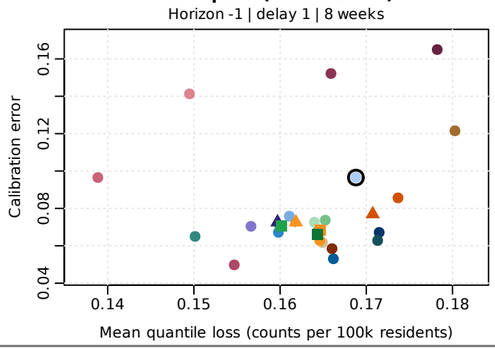
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks

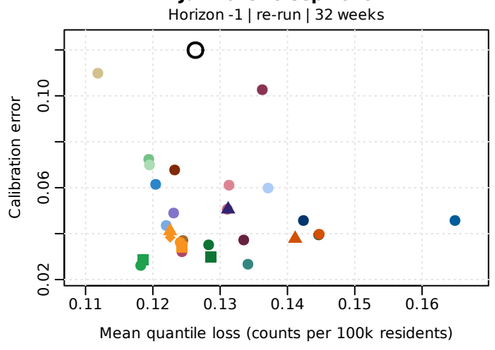


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

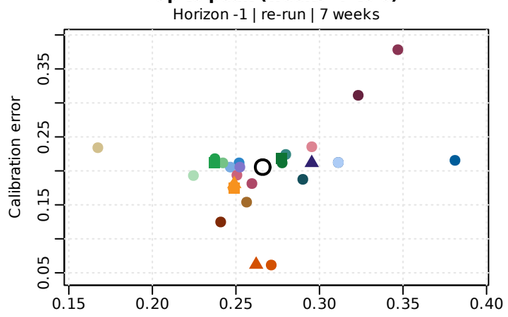


Feedback re-run

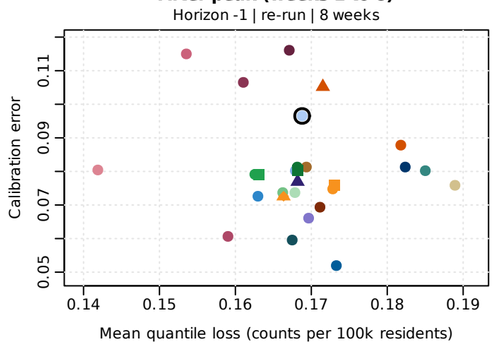
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks



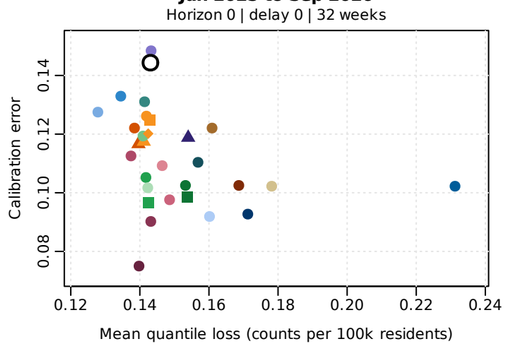
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



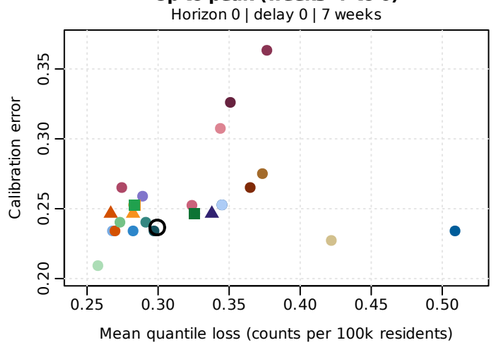
Horizon 0

Feedback delay 0

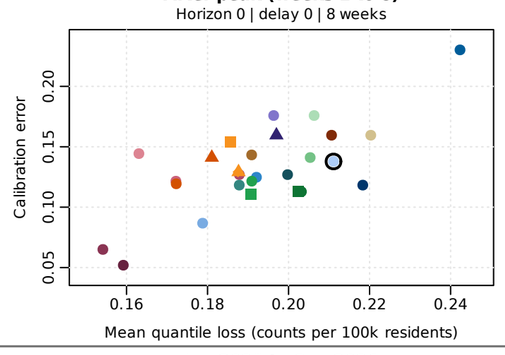
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks

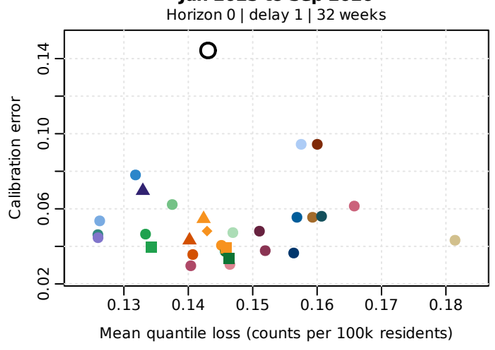


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

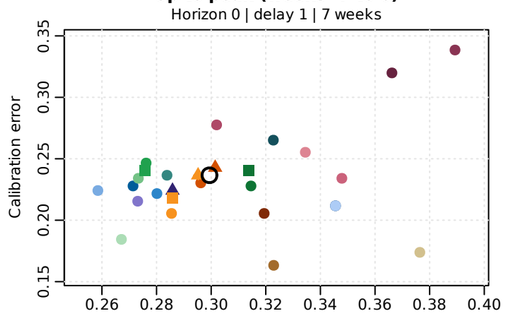


Feedback delay 1

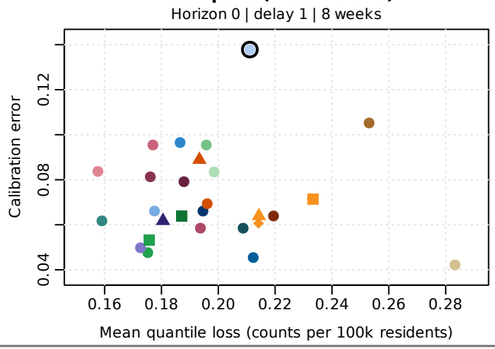
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks

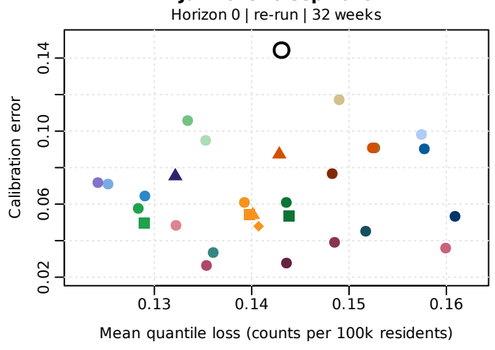


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

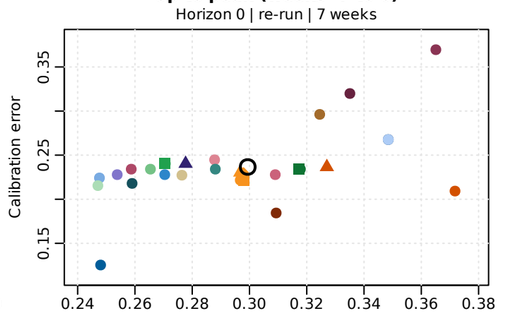


Feedback re-run

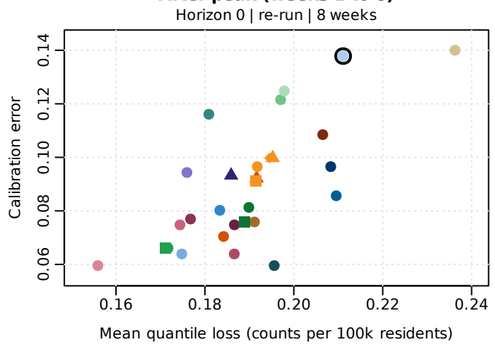
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks



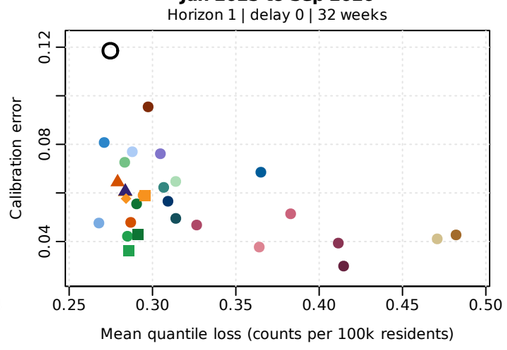
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



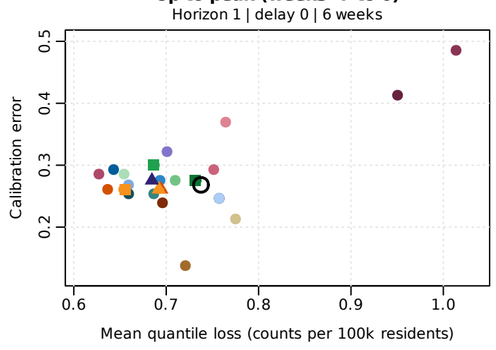
Horizon 1

Feedback delay 0

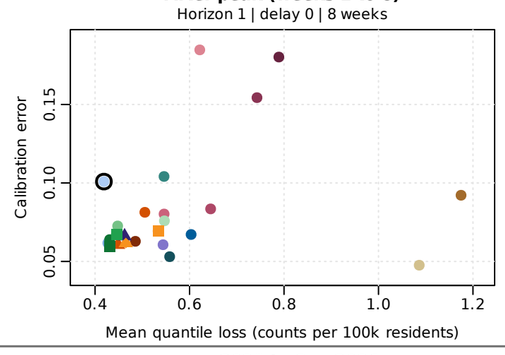
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks

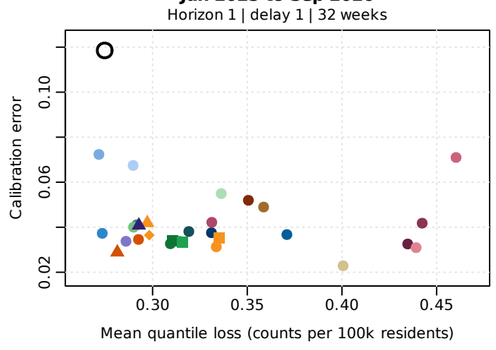


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

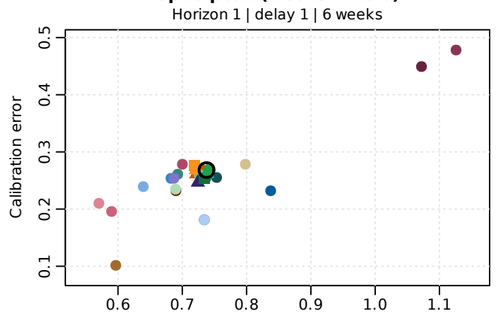


Feedback delay 1

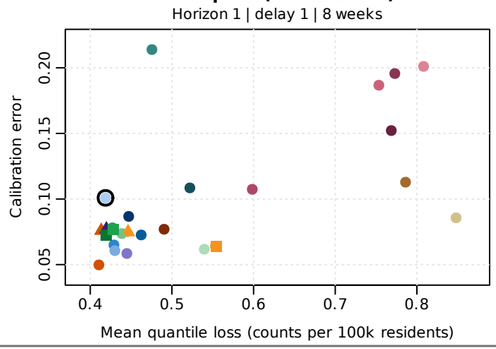
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks

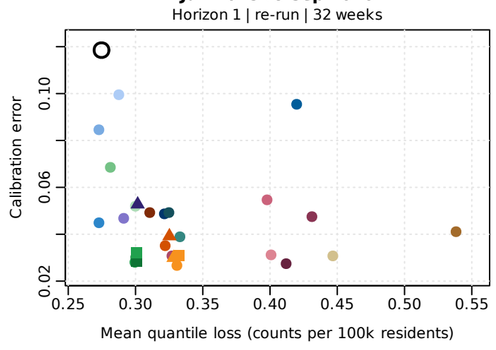


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

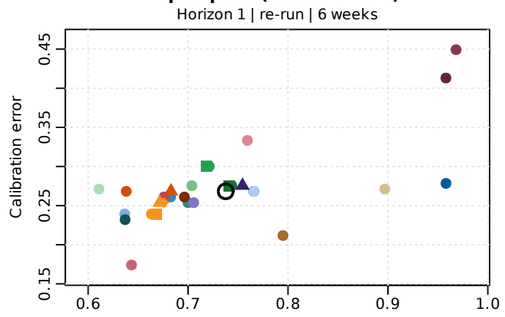


Feedback re-run

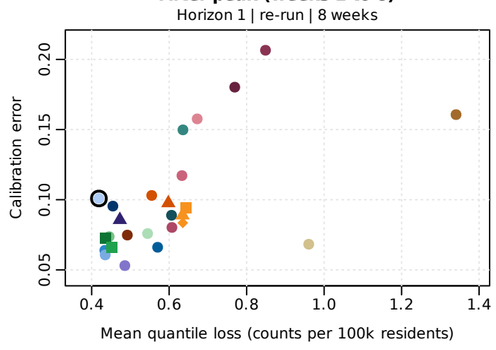
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks



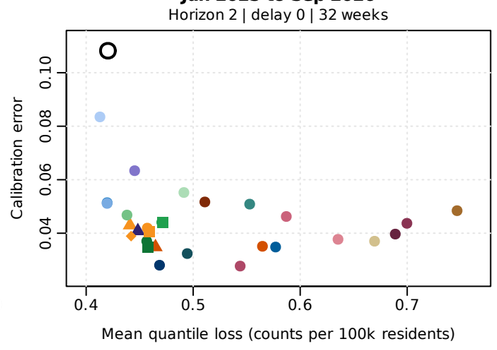
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



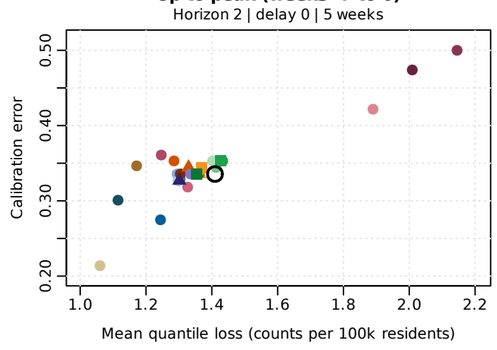
Horizon 2

Feedback delay 0

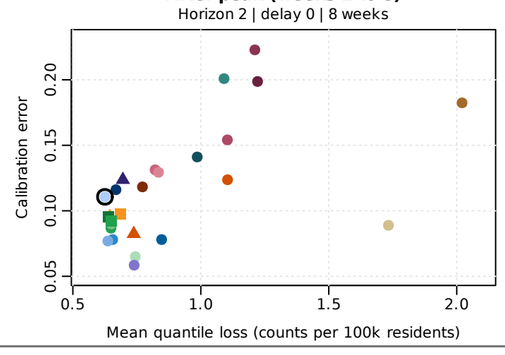
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks

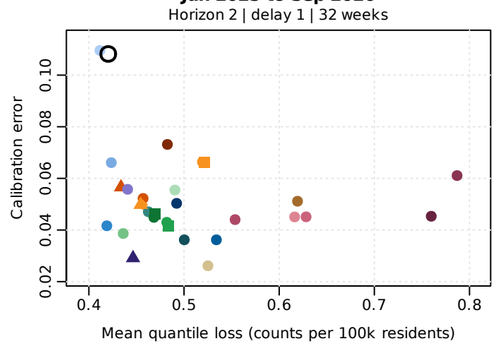


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

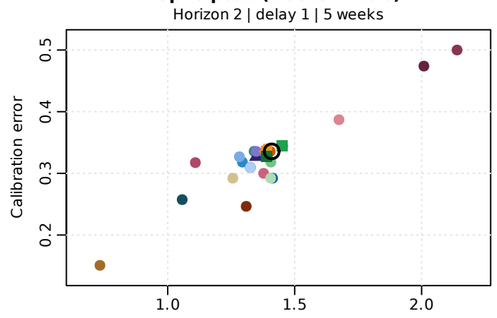


Feedback delay 1

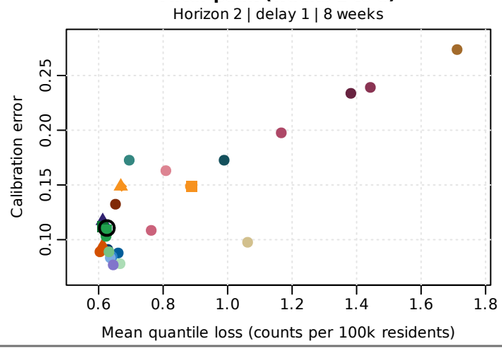
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks

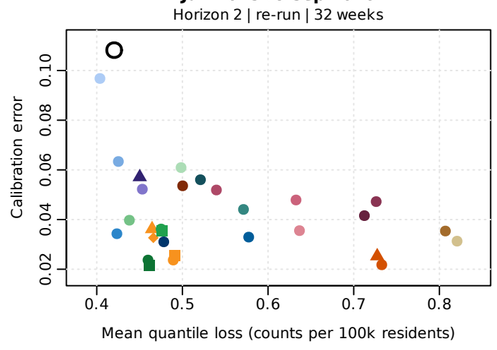


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

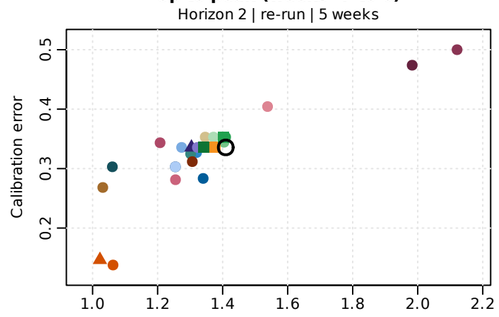


Feedback re-run

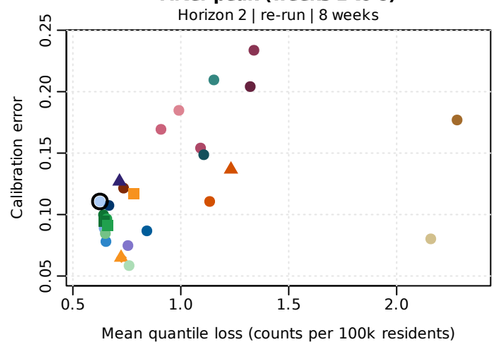
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks



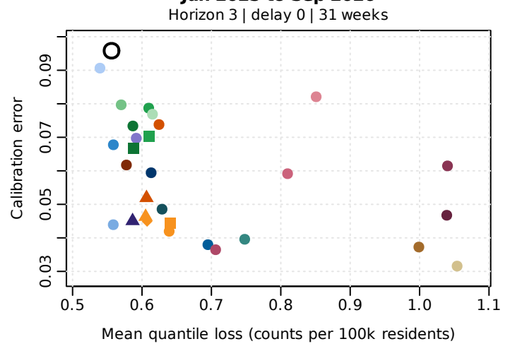
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



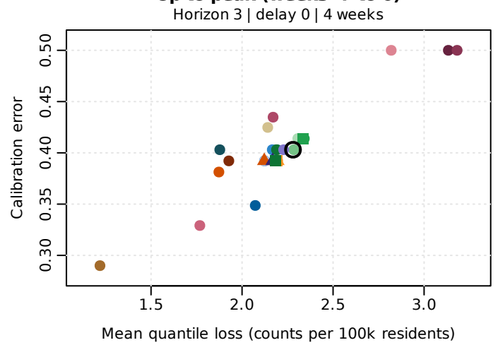
Horizon 3

Feedback delay 0

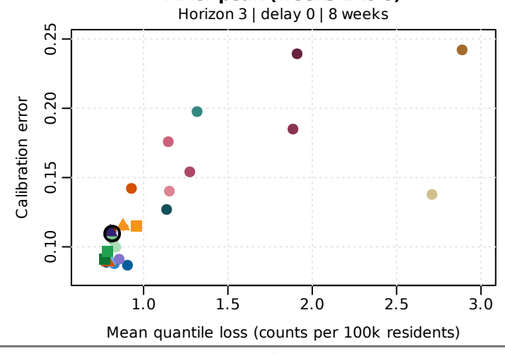
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks

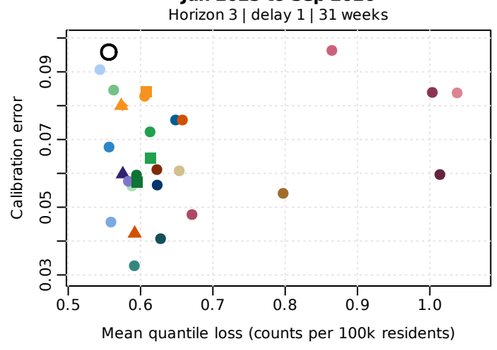


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

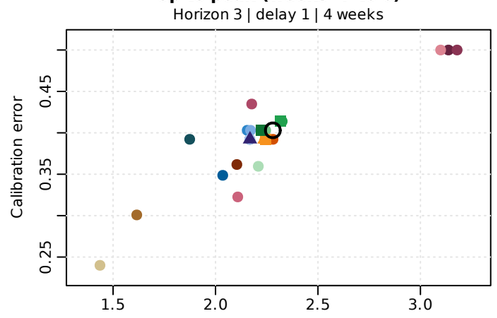


Feedback delay 1

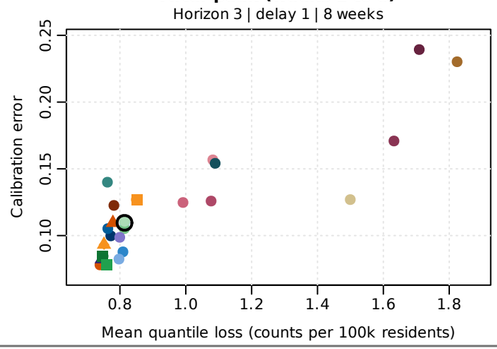
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks

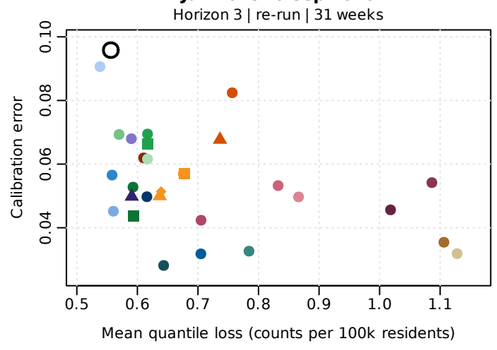


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks

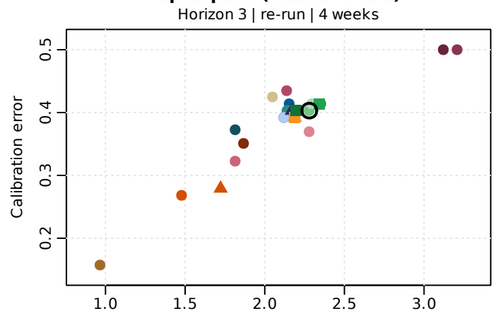


Feedback re-run

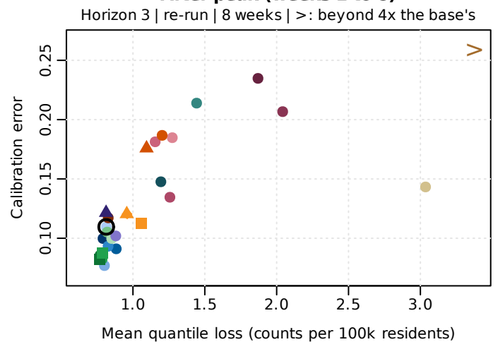
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks | >: beyond 4x the base's



Single trackers
- Base
- Fixed MultiQT (eta=0.05*S0)
- Leaky MultiQT (eta=0.2*S0, leak 0.1)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Target oracle (hindsight)

Log1p tracker
- Fixed MultiQT, log1p tracker (eta=0.2*S0)
- Leaky MultiQT, log1p tracker (eta=0.3*S0, leak 0.1)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=8)
- Rate grid: learned, no sharing, common weights (H=8)
- Rate grid: learned + sharing (H=8)
- Rate grid: learned + sharing, common weights (H=8)
- Rate grid: sharing + outer (H=8, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=8, outer m=0.001)

Weekly starts: immediate admission
- Weekly starts: equal (m=0.05)
- Weekly starts: learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
- Starts equal + outer (outer m=0.001)
- Starts learned + sharing + outer (H=8, outer m=0.001)

Score-based experts
- Score experts: equal weights
- Score experts: learned weights per level (H=8)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.3)
- Studentized median (RMS of 5, median start, eta=0.3)
- Studentized median (base 80% width, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, median start, eta=0.3)

Studentized full base
- Studentized full base (base 80% width, eta=0.3)
- Studentized full base (level RMS of 5, eta=0.3)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

flu | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

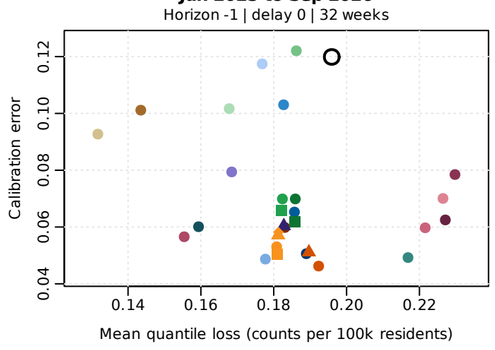
flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

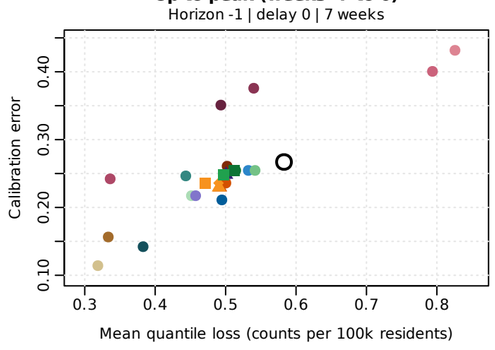
Feedback delay 0

Horizon -1

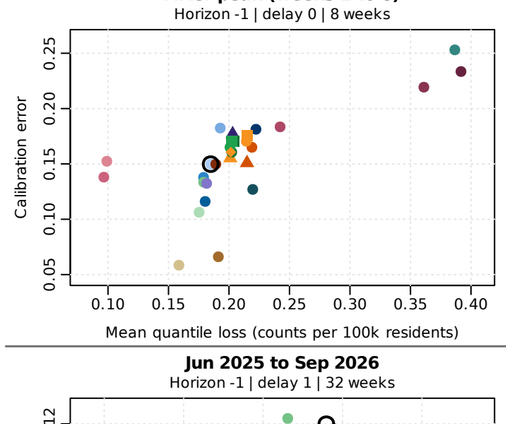
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks

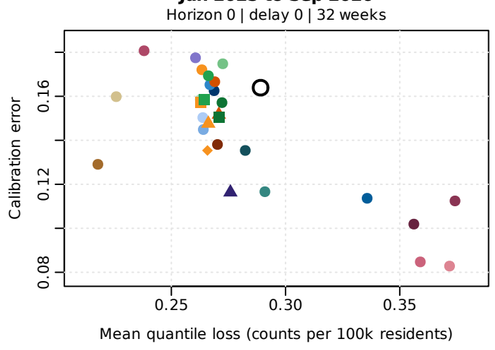


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

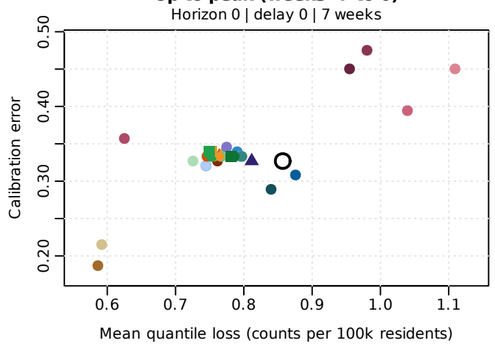


Horizon 0

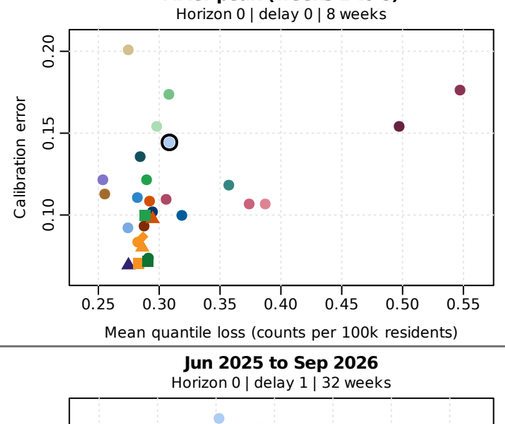
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks

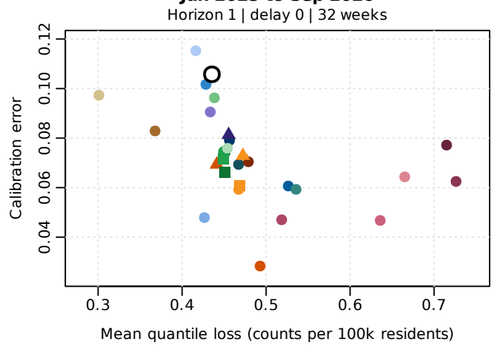


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

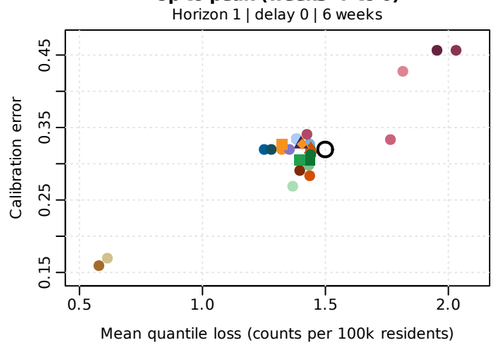


Horizon 1

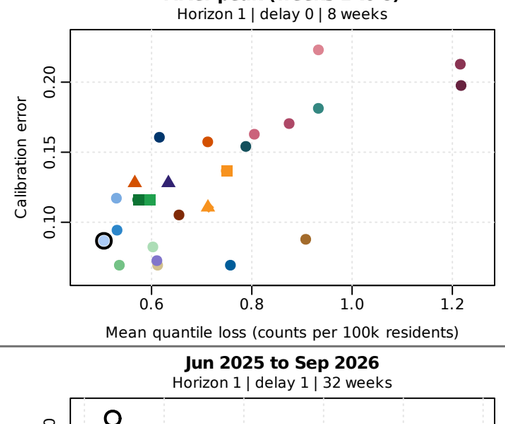
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks

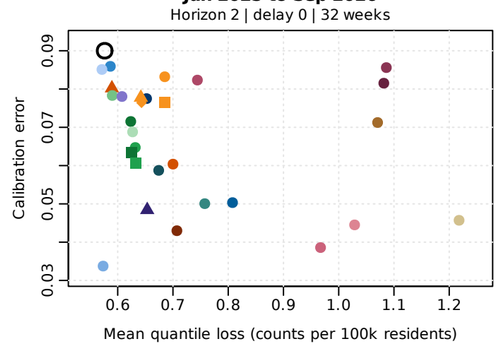


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

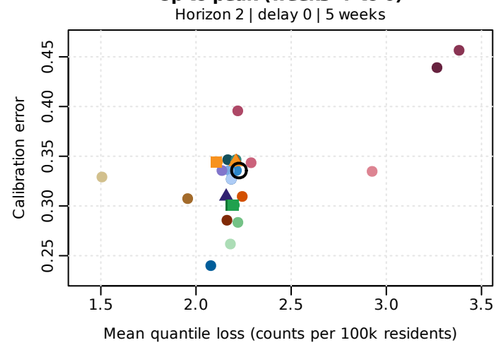


Horizon 2

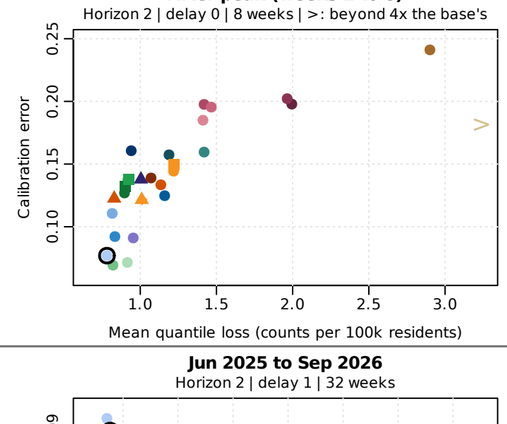
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks

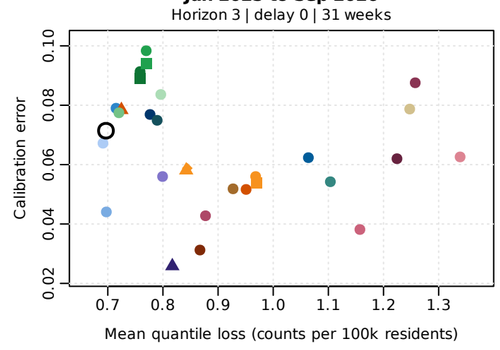


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks | >: beyond 4x the base's

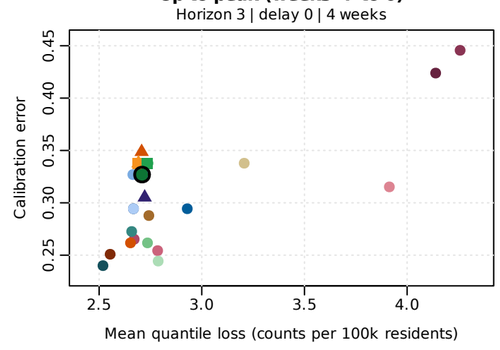


Horizon 3

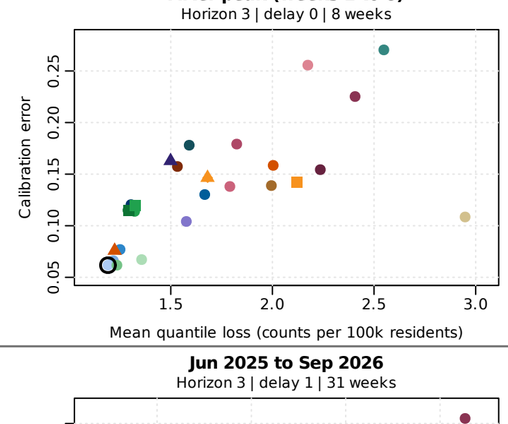
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks



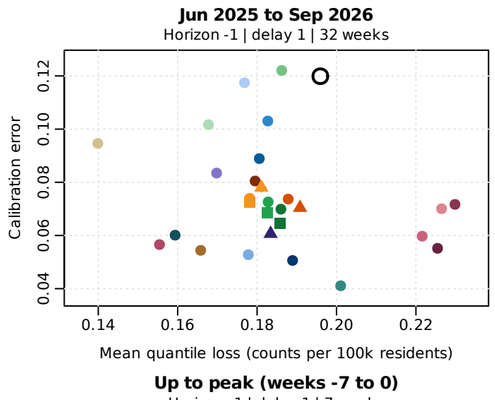
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks



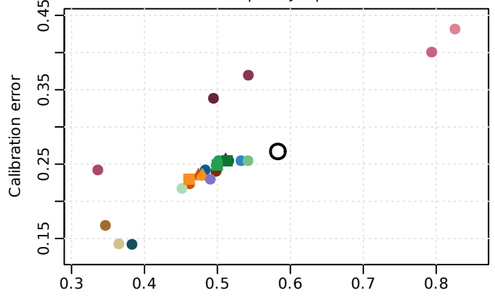
Feedback delay 1

Horizon -1

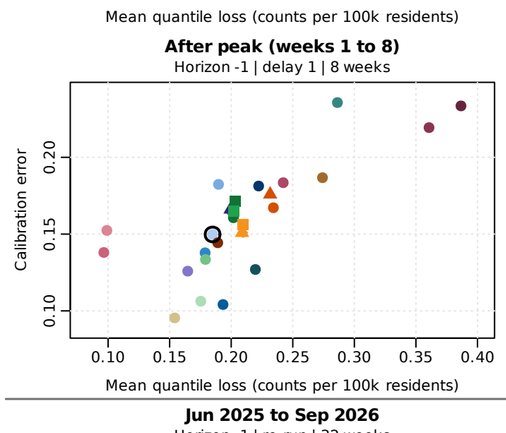
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks

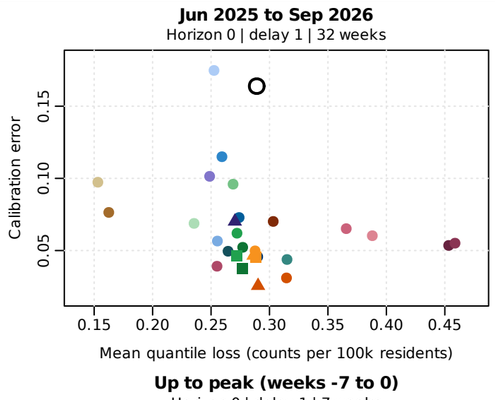


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

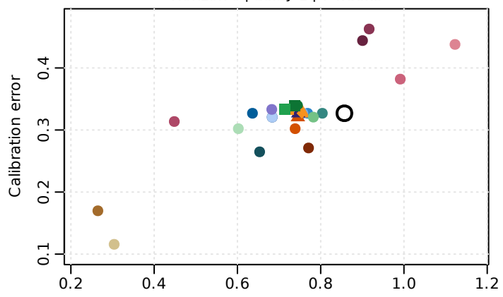


Horizon 0

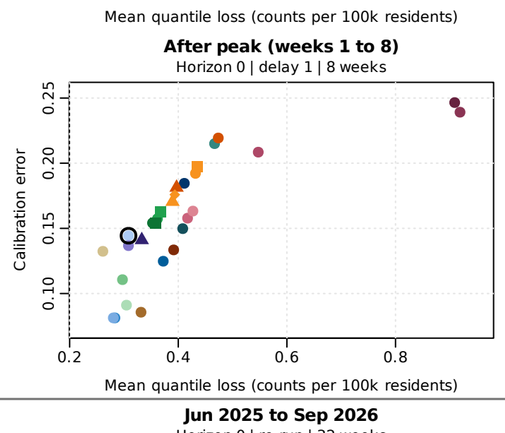
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks

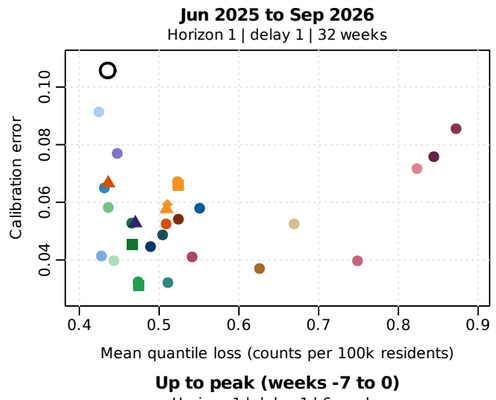


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

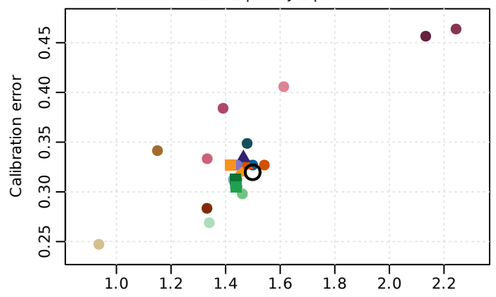


Horizon 1

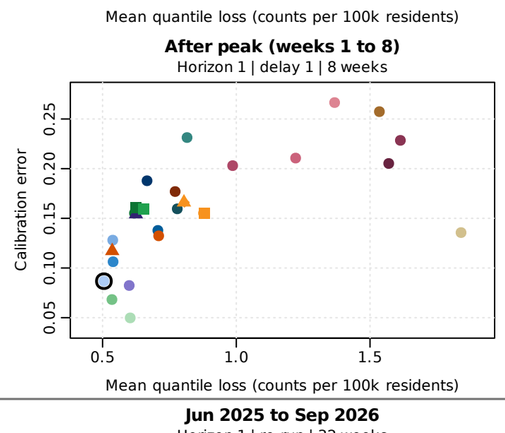
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks

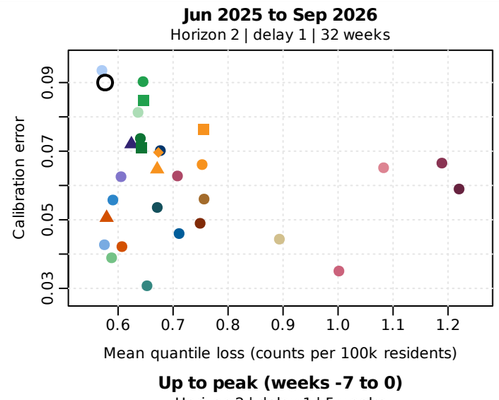


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

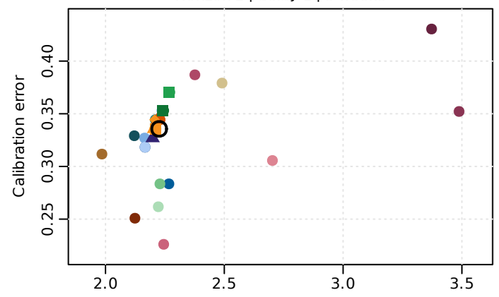


Horizon 2

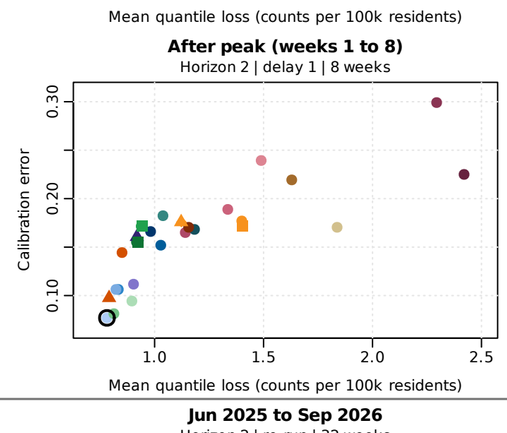
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks

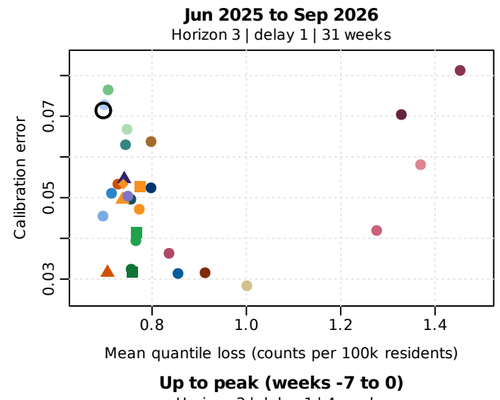


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

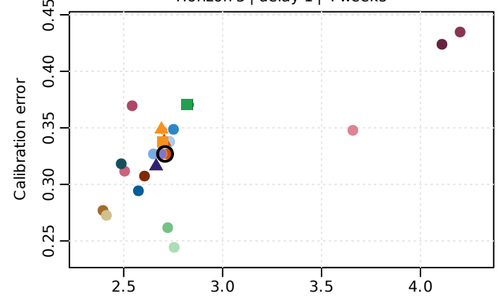


Horizon 3

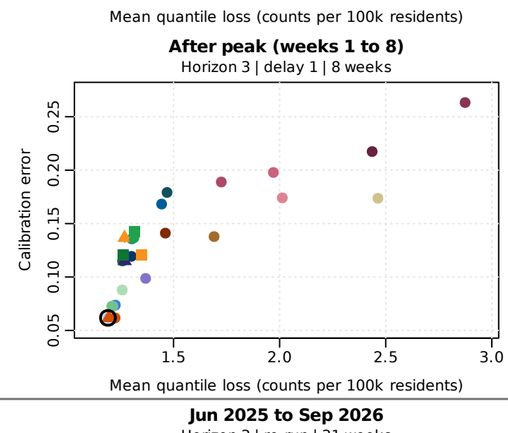
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks



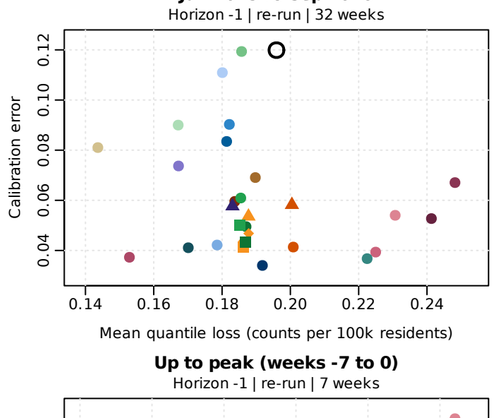
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks



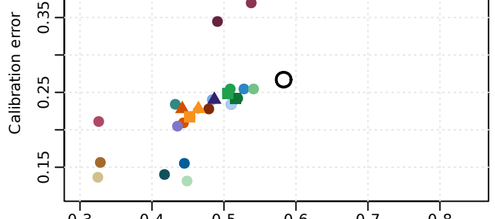
Feedback re-run

Horizon -1

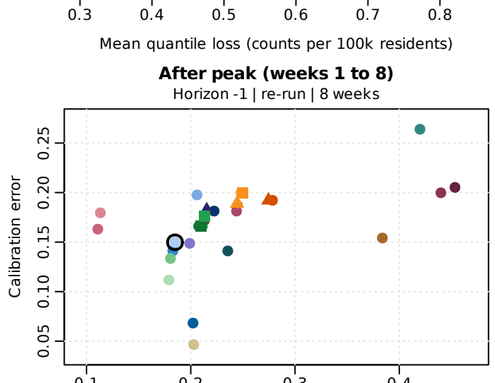
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks

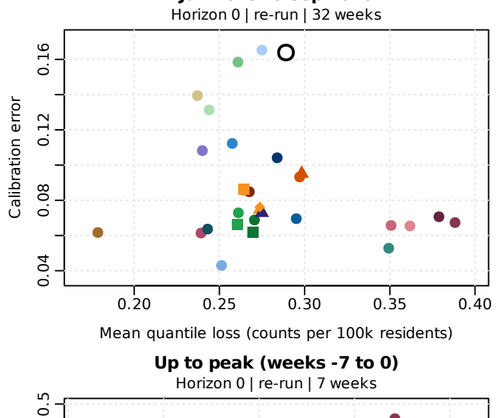


After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks

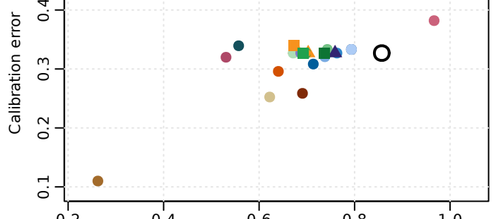


Horizon 0

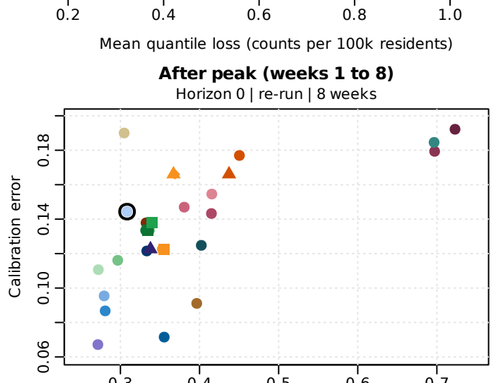
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks

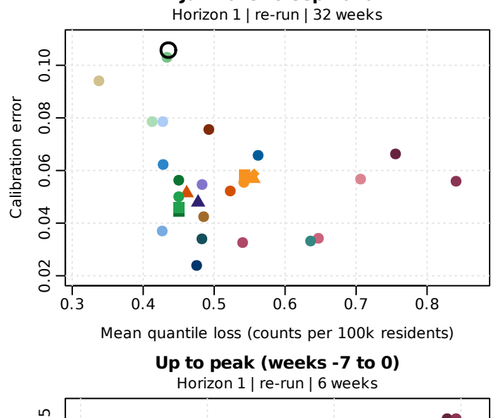


After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks

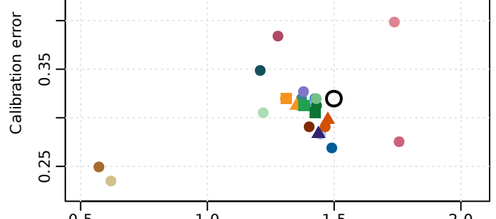


Horizon 1

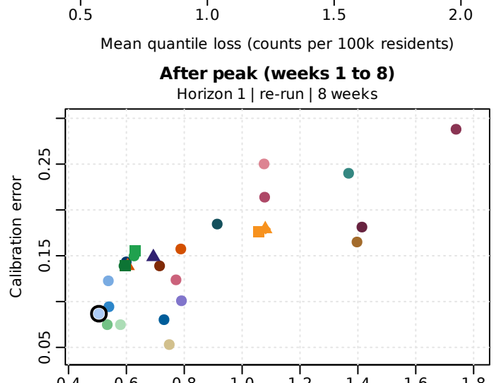
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks

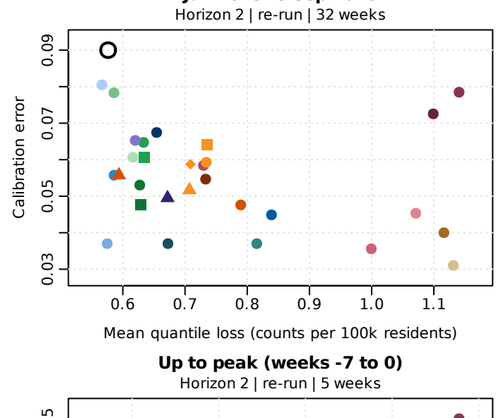


After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks

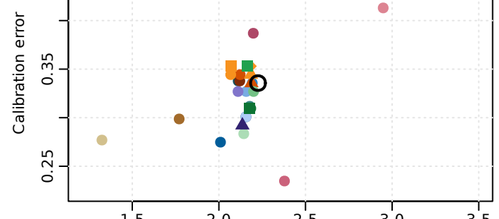


Horizon 2

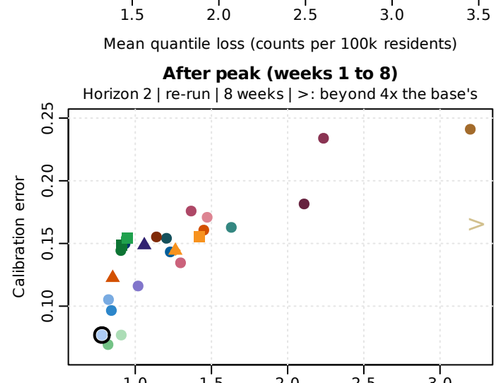
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks

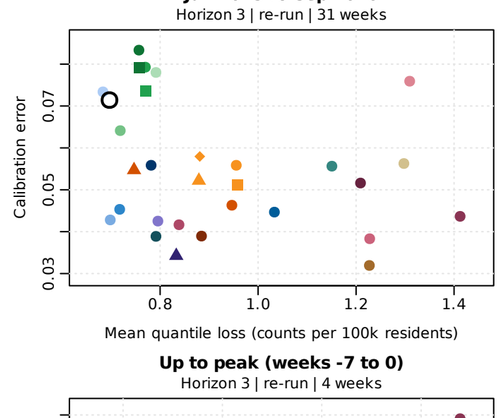


After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks | >: beyond 4x the base's

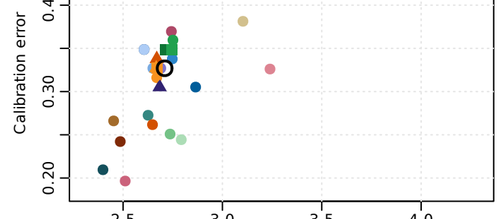


Horizon 3

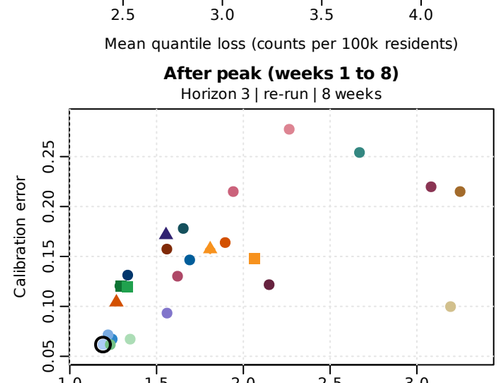
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks



Single trackers
- Base
- Fixed MultiQT (eta=0.05*S0)
- Leaky MultiQT (eta=0.2*S0, leak 0.1)
- Delphi REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- Delphi flu prod (sqrt per 100k, eta=0.018)
- Target oracle (hindsight)

Log1p tracker
- Fixed MultiQT, log1p tracker (eta=0.2*S0)
- Leaky MultiQT, log1p tracker (eta=0.3*S0, leak 0.1)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=8)
- Rate grid: learned, no sharing, common weights (H=8)
- Rate grid: learned + sharing (H=8)
- Rate grid: learned + sharing, common weights (H=8)
- Rate grid: sharing + outer (H=8, outer m=0.001)
- Rate grid: sharing, common weights + outer (H=8, outer m=0.001)

Weekly starts: immediate admission
- Weekly starts: equal (m=0.05)
- Weekly starts: learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, learned + sharing (m=0.05, H=8)
- Weekly starts: most recent 8, slow + fast, learned + sharing (m=0.03 + 0.3, H=8)
- Starts equal + outer (outer m=0.001)
- Starts learned + sharing + outer (H=8, outer m=0.001)

Score-based experts
- Score experts: equal weights
- Score experts: learned weights per level (H=8)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.3)
- Studentized median (RMS of 5, median start, eta=0.3)
- Studentized median (base 80% width, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, normal start, eta=0.3)
- Median, constant sigma (RMS of 5, median start, eta=0.3)

Studentized full base
- Studentized full base (base 80% width, eta=0.3)
- Studentized full base (level RMS of 5, eta=0.3)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).